

REPUBLIC OF RWANDA

EASTERN PROVINCE

KIREHE DISTRICT

KIGINA SECTOR

KIREHE ADVENTIST TECHNICAL SECONDARY SCHOOL

340 MACHINE LEARNING QUESTIONS AND ANSWERS

This document contains 340 machine learning questions covering various topics, including:

1. **Data Pre-processing** – Concepts of machine learning, data gathering, cleaning, visualization, and preparation.
2. **Developing ML Models** – Different ML algorithms, model training, selection, and evaluation.
3. **Model Deployment** – Deploying, integrating, and delivering ML predictions to clients.
4. **Advanced Topics** – Ethics, real-world applications, optimization, and technical aspects.

Let me hope that this document will be helpful to the Level 5 students in preparations of National Examinations.

Unit 1: Apply Data Pre-processing

1.1 Description of Machine Learning Concepts

1. What is machine learning?

- **English:** Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to learn from data and improve performance on tasks without being explicitly programmed.
- **Kinyarwanda:** Machine learning (ML) ni ishami rya **artificial intelligence (AI)** rigamije gufasha mudasobwa kwiga ku buryo bwikora **hifashishijwe amakuru** kugira ngo itere imbere mu mirimo itahawe amabwiriza by'umwihariko.

2. Define the term "artificial intelligence."

- **English:** Artificial Intelligence (AI) refers to the simulation of human intelligence in machines, enabling them to perform tasks that typically require human intelligence, such as problem-solving, decision-making, and pattern recognition.

- **Kinyarwanda: Artificial Intelligence (AI)** bivuze **gukoresha mudasobwa n'algorithm** kugira ngo bikore ibintu bisaba ubwenge bwa muntu nko gufata imyanzuro, gukemura ibibazo, no gutahura imiterere y'ibintu.

3. How is machine learning related to artificial intelligence?

- **English:** Machine learning is a subset of artificial intelligence. AI is the broader field that includes **rule-based systems and reasoning**, while machine learning focuses on **algorithms that learn from data** to make predictions and decisions.
- **Kinyarwanda: Machine Learning ni igice cya AI.** AI irimo uburyo bwose bwo gutuma mudasobwa ifata imyanzuro, naho **ML yibanda ku kwiga ku makuru** kugira ngo itange ibisubizo idahawe amabwiriza yose.

4. List the seven major steps in the machine learning life cycle.

- **English:**
 1. **Data Collection** – Gathering relevant data.
 2. **Data Preprocessing** – Cleaning and transforming data.
 3. **Feature Engineering** – Selecting useful features.
 4. **Model Selection** – Choosing an appropriate algorithm.
 5. **Model Training** – Teaching the model using training data.
 6. **Model Evaluation** – Assessing performance on test data.
 7. **Model Deployment & Maintenance** – Using and updating the model.
- **Kinyarwanda:**
 1. **Gutera amakuru (Data Collection)**
 2. **Gusukura no gutunganya amakuru (Data Preprocessing)**
 3. **Guhitamo ibintu bifasha (Feature Engineering)**
 4. **Guhitamo algorithm nziza (Model Selection)**
 5. **Gutoza mudasobwa (Model Training)**
 6. **Kugenzura uko ikora (Model Evaluation)**
 7. **Gushyira mu bikorwa no kuvugurura (Model Deployment & Maintenance)**

5. What is the purpose of the data gathering phase in the machine learning life cycle?

- **English:** Data gathering ensures that the machine learning model has **enough quality data** to learn from and make accurate predictions.
- **Kinyarwanda:** Gutera amakuru bigamije gutuma **mudasobwa igira amakuru ahagije** kugira ngo yige neza kandi itange ibisubizo bifatika.

6. Explain the difference between data preparation and data wrangling.

- **English:**
 - **Data Preparation** – General cleaning and formatting of data for analysis.
 - **Data Wrangling** – **Transforming and restructuring raw data** to make it suitable for a model.
- **Kinyarwanda:**

- **Data Preparation** – Gukora isuku rusange y’amakuru.
- **Data Wrangling** – Guhindura no kuvugurura amakuru atunganye neza ku buryo akoreshwa.

7. What are common issues found in real-world data?

- **English:** Missing values, duplicates, outliers, inconsistent formats, and biased data.
- **Kinyarwanda:** Ibibazo rusange ni **kubura amakuru, amakuru yikubiyemo, amakuru atajyanye, n’ibipimo bitajyanye n’ukuri.**

8. Why is data cleaning important before model training?

- **English:** It removes errors, fills missing values, and improves model accuracy.
- **Kinyarwanda:** Isuku y’amakuru ni **ingenzi** kugira ngo mudasobwa itore amakuru y’ukuri, atarimo amakosa.

9. Describe two analytical techniques used in data analysis.

- **English:**
 - **Descriptive Analytics** – Summarizing data (e.g., mean, median).
 - **Predictive Analytics** – Using past data to predict future trends.
- **Kinyarwanda:**
 - **Descriptive Analytics** – Gusobanura neza amakuru (nko kumenya impuzandengo).
 - **Predictive Analytics** – Gukoresha amakuru ya kera kugira ngo hamenyekane ibizaba imbere.

10. What is the goal of the model testing phase in the machine learning life cycle?

- **English:** To check the model’s accuracy and ensure it performs well on unseen data.
- **Kinyarwanda:** Kugenzura **neza niba algorithm ikora neza**, idafite ibibazo by’ubwibone (**overfitting**).

11. Explain what deployment means in the context of machine learning.

- **English:** Deployment is the process of making the trained ML model available for use in real-world applications.
- **Kinyarwanda:** **Deployment** ni **gushyira modeli yatojwe mu bikorwa**, ku buryo abantu cyangwa porogaramu bayikoreshe.

12. Give three real-world applications of machine learning.

- **English:**
 1. Fraud detection in banking
 2. Medical diagnosis
 3. Self-driving cars

- **Kinyarwanda:**
 1. Gupima uburiganya mu mabanki
 2. Gusuzuma indwara
 3. Imodoka zigenda zidafite abashoferi

Unit 1: Apply Data Pre-processing (Continued)

1.1 Description of Machine Learning Concepts (Continued)

13. How is image recognition used in machine learning applications?

- **English:**

Image recognition uses machine learning models to identify and classify objects, faces, or patterns in images. It is widely used in **facial recognition systems, medical imaging, and security surveillance.**
- **Kinyarwanda:**

Kumenya amashusho (image recognition) bikoresha machine learning kugira ngo imenye ibintu biri mu ishusho, amasura y'abantu, cyangwa ibimenyetso bitandukanye. Byifashishwa mu **suzuma ry'indwara, umutekano, no gutahura amasura.**

14. What is the role of machine learning in virtual personal assistants like Siri or Alexa?

- **English:**

Machine learning enables **Siri, Alexa, and Google Assistant** to **understand human language**, recognize voice patterns, and improve responses based on user interactions.
- **Kinyarwanda:**

Machine learning ifasha **Siri, Alexa na Google Assistant** kumva no gusesengura **amagambo y'abantu**, bityo bigashobora gusubiza neza abantu.

15. Describe one advantage and one disadvantage of using machine learning for fraud detection.

- **English:**
 - **Advantage:** It can detect fraud **faster and more accurately** than humans.
 - **Disadvantage:** It can sometimes **produce false positives**, flagging legitimate transactions as fraud.
- **Kinyarwanda:**
 - **Ibyiza:** Iratahura uburiganya **vuba kandi neza** kurusha abantu.
 - **Ibibi:** Hari igihe **ifata ubucuruzi bwemewe nk'uburiganya** (false positives).

16. What are some advantages of using machine learning in online shopping?

- **English:**
 - Personalized recommendations
 - Fraud detection
 - Improved customer experience through chatbots
- **Kinyarwanda:**
 - Gutanga **ibicuruzwa bijyanye n'ibyo umukiriya akunda**
 - Gutahura **uburiganya mu kwishyura**
 - Gufasha abakiriya binyuze muri **chatbots**

17. Explain the term 'trend analysis' in the context of machine learning.

- **English:**

Trend analysis refers to using **historical data** to identify **patterns and future trends** in areas like stock prices, sales, or customer behavior.
- **Kinyarwanda:**

Trend analysis bisobanura gusesengura amakuru ya kera kugira ngo **menyekane uko ibintu bizagenda mu gihe kizaza**, urugero nko mu **guhanga amoko mashya y'ibicuruzwa**.

18. What is the difference between machine learning and deep learning?

- **English:**
 - **Machine Learning** includes all algorithms that learn from data.
 - **Deep Learning** is a **subset of ML** that uses **neural networks** to learn complex patterns.
- **Kinyarwanda:**
 - **Machine Learning** ni urwego rusange rukoresha algorithms kugira ngo imyigire iboneke.
 - **Deep Learning** ni **igice cya ML** gikoresha **neural networks** kugira ngo gisesengure amakuru atandukanye.

19. How does supervised learning differ from unsupervised learning?

- **English:**
 - **Supervised Learning:** The model learns from **labeled data** (e.g., spam detection).
 - **Unsupervised Learning:** The model finds patterns in **unlabeled data** (e.g., customer segmentation).
- **Kinyarwanda:**

- **Supervised Learning:** Algorithm iriga hamwe n'ibisobanuro by'amakuru (nko kumenya e-mails za spam).
- **Unsupervised Learning:** Algorithm irikorera ubushakashatsi idafite ibisobanuro (nko gutandukanya abakiriya hashingiwe ku myitwarire yabo).

20. Give an example of a machine learning problem best solved with reinforcement learning.

- **English:**
 - **Self-driving cars** use reinforcement learning to **learn from trial and error** how to navigate roads.
- **Kinyarwanda:**
 - **Imodoka zigenda zidafite umushoferi** zikoresha reinforcement learning kugira ngo **zimenye uko zigenda mu mihanda** hifashishijwe **kwiga mu buryo bwa 'trial and error'**.

21. What are the main components of a machine learning algorithm?

- **English:**
 1. **Dataset** – Collection of data
 2. **Features** – Variables used for prediction
 3. **Algorithm** – Mathematical model
 4. **Loss Function** – Measures prediction errors
 5. **Optimizer** – Adjusts the model to improve accuracy
- **Kinyarwanda:**
 1. **Dataset** – Amakuru akoreshwa
 2. **Features** – Ibintu bigira uruhare mu iteganyagihe
 3. **Algorithm** – Modeli ibara ibisubizo
 4. **Loss Function** – Igipimo cy'amakosa
 5. **Optimizer** – Igira uruhare mu gukosora algorithm

22. How do machine learning models improve over time?

- **English:**
 - **They learn from new data** and adjust predictions based on feedback.
- **Kinyarwanda:**
 - **Ziga ku makuru mashya** kandi zigahindura imyanzuro hashingiwe ku feedback.

23. Name two industries where machine learning is commonly applied.

- **English:**
 - **Healthcare** (disease prediction)
 - **Finance** (fraud detection)
- **Kinyarwanda:**
 - **Ubuwuzi** (gutahura indwara)
 - **Ubukungu** (gutahura uburiganya mu mabanki)

24. What are the key differences between supervised, unsupervised, and semi-supervised learning?

- **English:**
 - **Supervised:** Labeled data
 - **Unsupervised:** No labels
 - **Semi-supervised:** Combination of both
- **Kinyarwanda:**
 - **Supervised:** Harimo ibisobanuro by'amakuru
 - **Unsupervised:** Nta bisobanuro bihari
 - **Semi-supervised:** Havanzemo ibice byombi

25. Which machine learning type would be best for spam email classification? Why?

- **English:**
 - **Supervised learning** is best because we have labeled emails (spam or not).
- **Kinyarwanda:**
 - **Supervised learning** niyo yiza kuko dufite **e-mails zamenyekanye ko ari spam**.

Unit 1: Apply Data Pre-processing (Continued)

1.1 Description of Machine Learning Concepts (Continued)

26. Describe the role of reinforcement learning in autonomous driving.

- **English:**

Reinforcement learning (RL) helps **self-driving cars** learn from their environment by **rewarding good decisions** (e.g., stopping at red lights) and **penalizing bad decisions** (e.g., running a red light). The car improves by trial and error.
- **Kinyarwanda:**

Reinforcement Learning (RL) ifasha **imodoka zitagira abashoferi** kwiga **uburyo bwo kugenda mu muhanda**. Iyo imodoka **ikora ibintu byiza (nko guhagarara ku itara ritukura)**, **irashimirwa**, ariko iyo **ikora amakosa (nko kurenga itara)**, **ihanwa**.

27. What is the purpose of model evaluation in the machine learning lifecycle?

- **English:**
 - Model evaluation **checks how well a trained model performs** on unseen data.
 - It ensures that the model is **accurate, generalizes well, and is not overfitting**.
- **Kinyarwanda:**
 - **Gusuzuma modeli** bigamije **kumenya niba ikora neza** ku makuru mashya.
 - **Bifasha kumenya niba modeli itanga ibisubizo nyabyo kandi itagendera ku makuru yatojwemo gusa**.

28. List three disadvantages of machine learning.

- **English:**
 1. **Data dependency** – ML models need a lot of data.
 2. **Computational cost** – Requires high processing power.
 3. **Bias and fairness issues** – ML can be biased if trained on biased data.
- **Kinyarwanda:**
 1. **Ikenera amakuru menshi** – ML ikora neza iyo ifite amakuru menshi.
 2. **Ikenera ubushobozi bukomeye** – Bisaba mudasobwa zikomeye.
 3. **Ishobora kugira ubusumbane** – ML ishobora gufata ibyemezo bidakwiye niba amakuru yarabayemo ivangura.

29. Explain how machine learning reduces human error in data analysis.

- **English:**
 - ML **automates data processing** and removes human biases, making analysis more accurate.
- **Kinyarwanda:**
 - Machine learning ituma amakosa akorwa n'abantu agabanuka, kuko yifashisha algorithms zidafite amarangamutima mu gusesengura amakuru.

30. What are common challenges faced when interpreting machine learning results?

- **English:**
 - **Lack of explainability** – Some models, like deep learning, act as "black boxes."
 - **Bias in data** – If the data is biased, the model will be biased too.
 - **Overfitting** – The model may work well in training but poorly in real-world data.
- **Kinyarwanda:**
 - **Kugira ibisubizo bisobanutse neza** – Hari igihe models zigira "black box" effect aho utamenya uko yafashe umwanzuro.
 - **Ivangura mu makuru** – Iyo amakuru yatojwemo arimo ubusumbane, model nayo izaba ifite ubusumbane.
 - **Overfitting** – Iyo model yiga cyane ku makuru yatojwemo, ishobora kudakora neza ku makuru mashya.

1.2 Preparing Machine Learning Environment

31. What is the first step in preparing a machine learning environment?

- **English:**
 - The first step is **installing a programming language** (like Python) and setting up necessary tools (IDE, libraries, frameworks).
- **Kinyarwanda:**

- **Intambwe ya mbere** ni gushyiraho ururimi rwa porogaramu (nko muri Python) no kwinjiza ibikoresho nka **libraries** na **frameworks**.

32. Name three popular Python IDEs used for machine learning.

- **English:**
 1. **Jupyter Notebook**
 2. **PyCharm**
 3. **VS Code**
- **Kinyarwanda:**
 1. **Jupyter Notebook**
 2. **PyCharm**
 3. **VS Code**

33. Describe the purpose of the Anaconda distribution in machine learning.

- **English:**
 - **Anaconda** is a Python distribution that comes with pre-installed ML libraries like **NumPy, Pandas, and TensorFlow**, making it easier to set up a data science environment.
- **Kinyarwanda:**
 - **Anaconda** ni igikoresho gikubiyemo **Python** na **libraries** z'ingenzi nka **NumPy, Pandas, na TensorFlow**, byorohereza abiga **machine learning**.

34. What is the role of a package manager like pip in Python?

- **English:**
 - **pip** is used to **install, update, and manage** Python libraries.
- **Kinyarwanda:**
 - **pip** ikoresheya mu **gushyiraho, kuvugurura no gucunga** libraries za Python.

35. List the basic hardware requirements for installing Python.

- **English:**
 - **At least 4GB RAM, a 64-bit processor, and a Windows, Mac, or Linux** operating system.
- **Kinyarwanda:**
 - **RAM ya 4GB nibura, processor ya 64-bit, hamwe na Windows, Mac, cyangwa Linux.**

36. Explain how to verify if Python has been installed successfully on a computer.

- **English:**
 - Open **Command Prompt (Windows)** or **Terminal (Mac/Linux)** and type:

```
python --version
```

- If Python is installed, it will display the version number.
- **Kinyarwanda:**
 - Fungura **Command Prompt (Windows)** cyangwa **Terminal (Mac/Linux)** wandike:

```
python --version
```

- Niba Python yarashizwemo, **izagaragaza version yayo**.

37. What is Jupyter Notebook, and why is it popular for machine learning projects?

- **English:**
 - **Jupyter Notebook** is an **interactive coding environment** that allows running Python code **cell by cell**, making it useful for **data analysis and visualization**.
- **Kinyarwanda:**
 - **Jupyter Notebook** ni **igikoresho cyo kwandika no gukora code** aho ushobora **kugabanya igice cya code** ukoresheje cells, bigafasha cyane mu **gusesengura no kwerekana amakuru**.

38. How do you install machine learning libraries like NumPy and Pandas using pip?

- **English:**
 - Open a terminal and type:

```
pip install numpy pandas
```

- **Kinyarwanda:**
 - Fungura terminal wandike:

```
pip install numpy pandas
```

Unit 1: Apply Data Pre-processing (Continued)

1.2 Preparing Machine Learning Environment (Continued)

39. What is the purpose of using cloud-based platforms like Google Colab?

- **English:**

- Google Colab provides a free cloud environment with **pre-installed machine learning libraries** like TensorFlow and allows users to run Python code **without installing anything on their local computer**.
- **Kinyarwanda:**
 - **Google Colab** ni urubuga **rutanga ubushobozi bwo gukora machine learning mu bushorishori (cloud)**. Rwifashishwa cyane kuko **rugira libraries zisanzwe zashyizwemo** kandi **ntibisaba kwishyiriraho software kuri mudasobwa yawe**.

40. Describe the steps to create a virtual environment using Anaconda.

- **English:**
 1. Open **Anaconda Prompt**.
 2. Create a new environment:

```
conda create --name myenv python=3.8
```

3. Activate the environment:

```
conda activate myenv
```

4. Install necessary libraries:

```
pip install numpy pandas
```

- **Kinyarwanda:**
 1. Fungura **Anaconda Prompt**.
 2. Tanga itegeko ryo gukora environment nshya:

```
conda create --name myenv python=3.8
```

3. Yemeza iyo environment:

```
conda activate myenv
```

4. Shyiramo libraries zikenewe:

```
pip install numpy pandas
```

41. Why is it important to install IDEs when preparing for machine learning development?

- **English:**
 - **IDEs** (Integrated Development Environments) like **Jupyter, PyCharm, and VS Code** provide tools to write, debug, and visualize code efficiently.
- **Kinyarwanda:**

- IDEs nka **Jupyter, PyCharm, na VS Code** zitanga uburyo bworoshye bwo kwandika no kugenzura amakosa ya code.

42. What does CI/CD stand for, and how is it related to machine learning workflows?

- **English:**
 - **CI/CD (Continuous Integration and Continuous Deployment)** ensures that ML models are automatically **tested, updated, and deployed** without manual intervention.
- **Kinyarwanda:**
 - **CI/CD (Continuous Integration and Continuous Deployment)** bifasha kugenzura, kuvugurura no gushyira mu bikorwa machine learning models mu buryo bwikora.

43. Explain the difference between TensorFlow and PyTorch.

- **English:**
 - **TensorFlow** is developed by Google and is widely used for **production ML models**.
 - **PyTorch** (by Facebook) is preferred for **research and prototyping** due to its **ease of debugging**.
- **Kinyarwanda:**
 - **TensorFlow** ni framework ya **Google**, ikoreshwa cyane mu gushyira mu bikorwa **models za machine learning**.
 - **PyTorch** ni framework ya **Facebook**, ikoreshwa cyane **muri ubushakashatsi no kugerageza prototypes** kubera **uburyo bworoshye bwo gukosora amakosa**.

44. What is hyperparameter tuning, and when is it performed in the ML process?

- **English:**
 - Hyperparameter tuning is the process of **optimizing model parameters** like learning rate and batch size to improve accuracy. It is performed **after training and evaluation**.
- **Kinyarwanda:**
 - **Hyperparameter tuning** ni uburyo bwo **guhindura neza imikorere ya model** (nko guhitamo **learning rate** na **batch size**) kugira ngo itange ibisubizo byiza. Ikorwa nyuma yo gutoza no kugenzura model.

45. Name two cloud services commonly used for machine learning model deployment.

- **English:**
 - **AWS SageMaker** – Amazon’s ML service for training and deploying models.

- **Google AI Platform** – Google’s ML service for scalable model deployment.
- **Kinyarwanda:**
 - **AWS SageMaker** – Serivisi ya Amazon ikoreshwa mu gutoza no gushyira models mu bikorwa.
 - **Google AI Platform** – Urubuga rwa Google rugenewe deployment ya models nini.

1.3 Data Collection and Acquisition

46. Define the term "dataset" in the context of machine learning.

- **English:**
 - A dataset is a structured collection of **data points** used to train, test, or evaluate a machine learning model.
- **Kinyarwanda:**
 - **Dataset** ni urutonde rw’amakuru akoreshwa mu **kwigisha, kugerageza, cyangwa kugenzura model**.

47. What are the six V’s of Big Data?

- **English:**
 1. **Volume** – Large amounts of data.
 2. **Velocity** – Speed at which data is generated.
 3. **Variety** – Different data types (text, images, etc.).
 4. **Veracity** – Trustworthiness of data.
 5. **Value** – How useful the data is.
 6. **Variability** – Changing nature of data.
- **Kinyarwanda:**
 1. **Volume** – Amakuru menshi cyane.
 2. **Velocity** – Umuvuduko w’ukuntu amakuru yinjira.
 3. **Variety** – Ubwoko butandukanye bw’amakuru.
 4. **Veracity** – icyizere abantu bagirira ayo makuru.
 5. **Value** – Akamaro ayo makuru afite.
 6. **Variability** – Uburyo amakuru ahinduka kenshi.

48. Differentiate between structured and unstructured data with examples.

- **English:**
 - **Structured Data:** Organized in rows and columns (e.g., a database table).
 - **Unstructured Data:** Lacks clear format (e.g., emails, images, videos).
- **Kinyarwanda:**
 - **Structured Data:** Amakuru **ateguwe neza** (nko muri **database**).
 - **Unstructured Data:** Amakuru **adasobanutse neza** (nko **amashusho, amajwi, cyangwa inyandiko**).

49. Explain the importance of data acquisition in machine learning.

- **English:**
 - Data acquisition ensures that models **have high-quality and relevant data**, improving performance.
- **Kinyarwanda:**
 - **Gutoranya amakuru** bifasha models kugira **amakuru meza, yizewe** kandi **ajyanye n'icyo yakozweho**, bikongera **ukuri k'ibisubizo**.

50. What is the role of IoT sensors in data collection for ML models?

- **English:**
 - IoT sensors **automatically collect real-time data** (e.g., temperature, motion) for machine learning applications.
- **Kinyarwanda:**
 - **IoT sensors** zitanga amakuru **mu buryo bwikora kandi mu gihe nyacyo** (nko kumenya **ubushyuhe cyangwa imikorere y'ibikoresho**).

Unit 1: Apply Data Pre-processing (Continued)

1.3 Data Collection and Acquisition (Continued)

51. Describe how smartphones can be used as a data source in machine learning.

- **English:**
 - Smartphones collect data from **sensors** (GPS, accelerometers, microphones) and **user interactions** (app usage, typing patterns) for ML applications like **health monitoring and personalized recommendations**.
- **Kinyarwanda:**
 - **Telefoni zigezweho** zitanga amakuru binyuze mu **dukoresho nka GPS, microfone, cyangwa accelerometer**, hamwe n'uburyo **abakoresha bakorana n'izo telefoni**. Ibi bikorwamo **ubuvuzi bwikoresha ikoranabuhanga n'inama zijyanye n'umuco wa buri mukoresha**.

52. What is a data warehouse, and how is it different from a traditional database?

- **English:**
 - **Data Warehouse:** Stores **historical and aggregated data** for analysis.
 - **Traditional Database:** Stores **real-time transactional data** for operations.
- **Kinyarwanda:**
 - **Data Warehouse:** Ibirundo by'amakuru **bikenewe mu gusesengura amakuru**.
 - **Database isanzwe:** Igira **amakuru y'igihe nyacyo** yifashishwa mu mikorere ya buri muni.

53. List three challenges faced during the data collection phase.

- **English:**
 1. **Incomplete data** – Missing values.
 2. **Data privacy issues** – Legal restrictions on data collection.
 3. **Inconsistent data formats** – Data from different sources may not match.
- **Kinyarwanda:**
 1. **Amakuru atuzuye** – Hari ibice by'amakuru bidahari.
 2. **Ibibazo by'amategako** – Hari aho gutoranya amakuru bitemewe.
 3. **Kutajyana kw'amakuru** – Iyo amakuru aturuka ahantu hatandukanye, ashobora kudahuza.

54. What is the difference between primary data and secondary data sources?

- **English:**
 - **Primary Data:** Collected directly from original sources (e.g., surveys).
 - **Secondary Data:** Data that was previously collected and used (e.g., reports).
- **Kinyarwanda:**
 - **Primary Data:** Amakuru yatoranijwe bwa mbere (nko gukora ubushakashatsi bwihariye).
 - **Secondary Data:** Amakuru yakusanyijwe mbere n'abandi (nko mu byegeranyo bya leta).

55. Explain the term "data labeling" in supervised machine learning.

- **English:**
 - **Data labeling** is the process of assigning **correct categories or tags** to data so that a machine learning model can learn patterns.
- **Kinyarwanda:**
 - **Data labeling** ni ukugenera amakuru ibisobanuro bizwi kugira ngo **machine learning** yige neza.

56. What are transactional data, and how are they relevant in machine learning?

- **English:**
 - **Transactional data** refers to records of business transactions (e.g., **bank transactions, online purchases**) and is used in **fraud detection and customer behavior prediction**.
- **Kinyarwanda:**
 - **Transactional data** ni amakuru ajyanye n'imyicarire y'ubucuruzi (nko amafaranga anyuzwa kuri konti, cyangwa ibigurwa kuri internet), akoresha mu gutahura uburiganya no kumenya imico y'abakiriya.

57. Describe the characteristics of Big Data.

- **English:**

- **Big Data** is described by **6 V's**:
 1. **Volume** – Large data amounts.
 2. **Velocity** – Fast generation.
 3. **Variety** – Different formats.
 4. **Veracity** – Data reliability.
 5. **Value** – Business impact.
 6. **Variability** – Changing nature.
- **Kinyarwanda:**
 - **Big Data** ifite ibiranga **6 V's**:
 1. **Volume** – Amakuru menshi.
 2. **Velocity** – Yihuta cyane.
 3. **Variety** – Ubwoko butandukanye.
 4. **Veracity** – Ukuri kw'amakuru.
 5. **Value** – Ibyo amariye ubucuruzi.
 6. **Variability** – Ahinduka cyane.

58. How can APIs be used to collect data for machine learning projects?

- **English:**
 - APIs (Application Programming Interfaces) allow developers to **fetch real-time data** from websites, databases, or IoT devices.
- **Kinyarwanda:**
 - APIs zituma abashakashatsi **babona amakuru mu buryo bwikora** baturutse kuri websites, databases, cyangwa ibikoreshe bya IoT.

59. What is data augmentation, and when is it applied in ML projects?

- **English:**
 - **Data augmentation** involves artificially increasing the dataset size by applying transformations (e.g., flipping images, adding noise). It is **used in deep learning for better model generalization**.
- **Kinyarwanda:**
 - **Data augmentation** ni ugukora impinduka ku makuru asanzwe (nko guhindura amashusho, kongeramo urusaku) kugira ngo **model yige neza** kandi idakora amakosa menshi.

60. Describe the purpose of using platforms like Kaggle for data acquisition.

- **English:**
 - **Kaggle** is an online platform that provides **datasets, competitions, and collaborative projects** for ML researchers.
- **Kinyarwanda:**
 - **Kaggle** ni urubuga rutanga **datasets zitandukanye, amarushanwa, n'uburyo bwo gufatanya ku mishinga ya ML**.

Unit 1: Apply Data Pre-processing (Continued)

1.4 Interpret Data Visualization

61. What is data visualization, and why is it important in machine learning?

- **English:**
 - **Data visualization** is the process of representing data **graphically** using charts, graphs, and plots.
 - It helps in **understanding patterns, trends, and insights** in data.
- **Kinyarwanda:**
 - **Data visualization** ni uburyo bwo kwerekana amakuru mu buryo bugaragara hifashishijwe **amaseti (graphs)**, **ibishushanyo**, **cyangwa imbata (plots)**.
 - Ifasha kumva imiterere y'amakuru no kubona aho ahuriye.

62. Name three popular data visualization libraries in Python.

- **English:**
 1. **Matplotlib**
 2. **Seaborn**
 3. **Plotly**
- **Kinyarwanda:**
 1. **Matplotlib**
 2. **Seaborn**
 3. **Plotly**

63. What is Matplotlib, and how is it used in data visualization?

- **English:**
 - **Matplotlib** is a Python library used to create **static, animated, and interactive visualizations**.
 - Example usage:

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4])
plt.show()
```

- **Kinyarwanda:**
 - **Matplotlib** ni library ya Python ikoreshwa mu gukora ibishushanyo by'amakuru.
 - Urugero rw'ikoreshwa ryayo:

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4])
plt.show()
```

64. Explain the difference between Matplotlib and Seaborn.

- **English:**
 - **Matplotlib:** Provides **basic** plotting functions.
 - **Seaborn:** Built on top of Matplotlib, offering **more advanced and attractive visualizations**.
- **Kinyarwanda:**
 - **Matplotlib:** Irimo uburyo **busanzwe bwo gushushanya**.
 - **Seaborn:** Ikora **ibishushanyo byinshi byimbitse kandi binogeye ijisho**.

65. How does Plotly differ from traditional static visualization tools?

- **English:**
 - **Plotly** creates **interactive visualizations**, allowing users to zoom, hover, and explore data dynamically.
- **Kinyarwanda:**
 - **Plotly** ikora **ibishushanyo by'amakuru bifite ubusabane**, aho ushobora **kugenda mu ishusho, kongera cyangwa kugabanya**.

66. What type of data is best visualized using a scatter plot?

- **English:**
 - A **scatter plot** is used to show **relationships between two numerical variables**.
- **Kinyarwanda:**
 - **Scatter plot** ikoresheya mu **kugaragaza isano iri hagati y'inyuguti ebyiri z'imibare**.

67. When would you use a heatmap in data analysis?

- **English:**
 - **Heatmaps** are used to show **correlations and intensity differences** between data points.
- **Kinyarwanda:**
 - **Heatmap** ikoresheya mu **kugaragaza uko ibintu bihuriye hamwe no kwerekana uburemere bw'amakuru**.

68. Describe the purpose of using dashboards like Tableau or Power BI.

- **English:**
 - Dashboards provide **real-time interactive visualizations** for monitoring business and ML models.
- **Kinyarwanda:**
 - **Dashboards** nka **Tableau cyangwa Power BI** zitanga **uburyo bwo kugenzura amakuru mu buryo burambuye kandi bwihuse**.

69. What is the role of data visualization in identifying data anomalies?

- **English:**

- **Data visualization** helps in detecting outliers, missing values, and inconsistencies **more quickly** than raw data.
- **Kinyarwanda:**
 - **Gushushanya amakuru** bifasha gutahura amakosa y'amakuru, ibice bituzuye, cyangwa amakuru adahuye.

70. Explain how a histogram helps in understanding data distribution.

- **English:**
 - **Histograms** show **how data is spread out**, helping analysts identify skewness and outliers.
- **Kinyarwanda:**
 - **Histogram** yerekana **uko amakuru akwirakwiriye** kandi **ifasha kumenya niba hari amakuru arimo gutwarwa ku ruhande rumwe**.

71. What are box plots used for in data visualization?

- **English:**
 - Box plots help detect **outliers** and display **median, quartiles, and range** in numerical data.
- **Kinyarwanda:**
 - **Box plot** ikoresheya mu **kugaragaza uko imibare igabanijemo ibice no gutahura amakosa (outliers)**.

72. Describe the process of creating a bar chart using Matplotlib.

- **English:**
 - Example code:

```
import matplotlib.pyplot as plt
categories = ["A", "B", "C"]
values = [10, 20, 15]
plt.bar(categories, values)
plt.show()
```

- **Kinyarwanda:**
 - Urugero rwa **bar chart** muri **Matplotlib**:

```
import matplotlib.pyplot as plt
categories = ["A", "B", "C"]
values = [10, 20, 15]
plt.bar(categories, values)
plt.show()
```

73. What is the importance of labeling axes in data visualization?

- **English:**
 - Labeled axes help users **understand what the graph represents.**
- **Kinyarwanda:**
 - **Labeling** bifasha abantu kumva icyo ishusho y'amakuru ivuga.

74. How can visualizations help in feature selection for machine learning models?

- **English:**
 - Visualizations help **identify which features have the most impact** on predictions.
- **Kinyarwanda:**
 - **Ibishushanyo** by'amakuru bifasha kumenya ibihe bintu bifite akamaro mu gutanga ibisubizo byiza.

75. Explain the concept of correlation matrices in data visualization.

- **English:**
 - A **correlation matrix** shows how **features relate to each other**, helping in feature selection.
- **Kinyarwanda:**
 - **Correlation matrix** yerekana **uko ibintu bihuriye hamwe**, bigafasha guhitamo ibifite agaciro.

Unit 1: Apply Data Pre-processing (Continued)

1.5 Perform Data Cleaning

76. What is data cleaning in machine learning?

- **English:**
 - **Data cleaning** is the process of **removing errors, handling missing values, and fixing inconsistencies** in data to improve model performance.
- **Kinyarwanda:**
 - **Data cleaning** ni uburyo **bwo gukosora amakosa, kwuzuza amakuru abuze, no gukemura ibibazo by'ubusumbane** kugira ngo model ikore neza.

77. List three common data quality issues addressed during data cleaning.

- **English:**
 1. **Missing values** – Some data points are missing.

2. **Duplicate data** – The same records appear multiple times.
3. **Outliers** – Extreme values that may distort analysis.
- **Kinyarwanda:**
 1. **Amakuru abuze** – Hari aho ibintu bimwe bituzuye.
 2. **Amakuru yikubiyemo** – Hari aho ibintu bisa bikubiyemo incuro nyinshi.
 3. **Outliers** – Hari aho imibare ibangamye igaragara mu makuru.

78. Explain the term "missing data" and how it can affect machine learning models.

- **English:**
 - **Missing data** refers to **incomplete records** in a dataset.
 - It can cause **biased predictions and reduced accuracy** in ML models.
- **Kinyarwanda:**
 - **Missing data** ni **amakuru atuzuye**.
 - Ashobora gutuma **ibisubizo by'algorithm bidahuye n'ukuri**, bikagabanya ubunyamwuga bwa model.

79. What is data imputation, and when is it necessary?

- **English:**
 - **Data imputation** is replacing missing values with estimates (e.g., mean, median).
 - It is necessary when **data is incomplete but still useful**.
- **Kinyarwanda:**
 - **Data imputation** ni **gusimbuza amakuru abuze ibipimo bigereranywa** (nko **impuzandengo cyangwa agaciro k'ahafi**).
 - Ikenewe **niba amakuru atuzuye ariko agifite akamaro**.

80. Describe two methods for handling missing values in datasets.

- **English:**
 1. **Remove missing values** – Delete rows/columns with too many missing values.
 2. **Imputation** – Fill missing values with **mean, median, or mode**.
- **Kinyarwanda:**
 1. **Gukuraho amakuru abuze** – Gusiba imyirondoro ifite amakuru make cyane.
 2. **Kuzuza aho habuze** – Gushyiramo **impuzandengo, agaciro gasanzwe** (median), cyangwa **igikunze kugaragara** (mode).

81. What is the difference between duplicate data and inconsistent data?

- **English:**
 - **Duplicate data**: The same records appear more than once.
 - **Inconsistent data**: Different formats or conflicting values in the dataset.
- **Kinyarwanda:**
 - **Duplicate data**: Iyo **amakuru yikubiyemo incuro nyinshi**.

- **Inconsistent data:** Iyo amakuru afite ibibazo mu miterere yayo, nk'aho igipimo kimwe gifite uburyo butandukanye bwo kwandikwa.
-

82. How can outliers negatively impact machine learning models?

- **English:**
 - **Outliers** can skew results, reduce model accuracy, and make training difficult.
- **Kinyarwanda:**
 - **Outliers** zituma ibisubizo bya model bihindagurika, bigatuma itanga ibipimo bidahuye n'ukuri.

83. What techniques are used to detect outliers in data?

- **English:**
 1. **Box Plots** – Graphical method to detect extreme values.
 2. **Z-Score** – Measures how far a data point is from the mean.
 3. **IQR (Interquartile Range)** – Identifies values outside expected range.
- **Kinyarwanda:**
 1. **Box Plots** – Gushushanya ibintu by'ingenzi kugira ngo turebe ibintu bidahuye n'ibisanzwe.
 2. **Z-Score** – Gupima ukuntu agaciro kari kure ugereranyije n'andi.
 3. **IQR** – Gusuzuma niba agaciro kanini cyangwa gato katandukanye cyane n'ibindi.

84. Explain the importance of data normalization in machine learning.

- **English:**
 - **Normalization** scales numerical values to a common range, improving model performance.
- **Kinyarwanda:**
 - **Normalization** ni ugushyira imibare ku gipimo gihwanye, bigatuma model ikora neza.

85. What is data transformation, and why is it done during data cleaning?

- **English:**
 - **Data transformation** is converting raw data into a structured format, making it easier for ML models to learn.
- **Kinyarwanda:**
 - **Data transformation** ni uguhindura amakuru atari mu mwirondoro wihariye kugira ngo akoreshwe neza.

86. How do you handle categorical data in machine learning preprocessing?

- **English:**

- Categorical data is transformed into numbers using:
 - **Label Encoding** – Assigning numbers to categories.
 - **One-Hot Encoding** – Creating binary columns for each category.
- **Kinyarwanda:**
 - Amakuru afite ibyiciro **ahindurwamo imibare** ukoresheje:
 - **Label Encoding** – Gutanga imibare kuri buri cyiciro.
 - **One-Hot Encoding** – Gukora **imyirondoro myinshi yerekana ibyiciro binyuranye**.

87. Describe the process of feature scaling and why it's important.

- **English:**
 - Feature scaling ensures that numerical features are in the same range, preventing bias in ML models.
- **Kinyarwanda:**
 - **Feature scaling** ifasha gushyira imibare ku gipimo gihwanye, bikarinda ko model ifata ibyemezo idashingiye ku ngingo zose.

88. What are the potential risks of over-cleaning a dataset?

- **English:**
 - **Over-cleaning** can remove valuable information and distort real-world patterns.
- **Kinyarwanda:**
 - **Over-cleaning** ishobora **gukuraho amakuru akenewe**, bigatuma model idatanga ibisubizo bifite ishingiro.

89. Explain the concept of data leakage and how to prevent it.

- **English:**
 - **Data leakage** occurs when training data contains information from test data, leading to **overfitting**.
 - **Prevention:** Separate training and test data correctly.
- **Kinyarwanda:**
 - **Data leakage** iba iyo amakuru y'igerageza yihishe mu y'imyitozo, bikayobya model.
 - **Uburyo bwo kuyirinda:** Gukora imiyoboro myiza igabanya amakuru mu matsinda atandukanye.

Unit 2: Develop Machine Learning Model

2.1 Description of Machine Learning Algorithms and Applications

90. What is a machine learning algorithm?

- **English:**
 - A **machine learning algorithm** is a mathematical procedure that learns patterns from data to make predictions or decisions.
- **Kinyarwanda:**
 - Machine learning algorithm ni uburyo bwa siyansi bwo kwiga ku makuru no gufata imyanzuro.

91. List three types of machine learning algorithms.

- **English:**
 1. **Supervised Learning** (e.g., Decision Trees, SVM)
 2. **Unsupervised Learning** (e.g., K-Means, PCA)
 3. **Reinforcement Learning** (e.g., Q-Learning)
- **Kinyarwanda:**
 1. **Supervised Learning** (nko kuri Decision Trees, SVM)
 2. **Unsupervised Learning** (nko kuri K-Means, PCA)
 3. **Reinforcement Learning** (nko kuri Q-Learning)

92. Explain the difference between classification and regression algorithms.

- **English:**
 - **Classification:** Predicts **categories** (e.g., spam vs. non-spam emails).
 - **Regression:** Predicts **continuous values** (e.g., house prices).
- **Kinyarwanda:**
 - **Classification:** Igira ibyiciro (e.g., email ni spam cyangwa si spam).
 - **Regression:** Itanga umusaruro uhoraho (e.g., kumenya igiciro cy'inzu).

93. What is the main objective of clustering algorithms?

- **English:**
 - Clustering algorithms **group similar data points together** without predefined labels.
- **Kinyarwanda:**
 - Clustering algorithm **itandukanya amakuru akurikije ibisa hagati yayo** hatabayeho igisobanuro cyihariye.

94. Describe how decision trees work in machine learning.

- **English:**
 - A **decision tree** splits data into smaller subgroups based on decision rules until it reaches a final classification or prediction.
- **Kinyarwanda:**
 - **Decision tree** itandukanya amakuru **hifashishijwe amabwiriza akurikiranye**, kugeza ubwo igenzura ryayo rirangiye.

95. What are the advantages of using Random Forest over a single decision tree?

- **English:**
 - **Random Forest** combines multiple decision trees, improving **accuracy and reducing overfitting**.
- **Kinyarwanda:**
 - **Random Forest** ihuza **decision trees nyinshi**, bikayifasha **kugira ubunyamwuga no kwirinda kwiga cyane ku makuru (overfitting)**.

96. Define the term "support vector machine" (SVM).

- **English:**
 - **SVM** is an algorithm that finds the **optimal boundary** that best separates two classes.
- **Kinyarwanda:**
 - **SVM** ni **algorithm ifasha gutandukanya ibyiciro bibiri hifashishijwe umurongo mwiza ushoboka**.

97. In which scenario would you prefer to use logistic regression over linear regression?

- **English:**
 - **Logistic Regression** is used for **classification problems** (e.g., predicting whether a customer will buy a product).
 - **Linear Regression** is used for **continuous predictions** (e.g., predicting sales revenue).
- **Kinyarwanda:**
 - **Logistic Regression** ikoreshwa **mu byiciro** (e.g., kumenya niba umukiriya azagura igicuruzwa cyangwa atazakigura).
 - **Linear Regression** ikoreshwa **mu mibare ihoraho** (e.g., guteganya igihombo cyangwa inyungu).

98. What is the purpose of using K-Nearest Neighbors (KNN) in machine learning?

- **English:**
 - **KNN** classifies a data point based on the majority class of its closest neighbors.
- **Kinyarwanda:**
 - **KNN** igena ibyiciro hashingiwe ku makuru aherereye hafi yayo.

99. Explain the concept of overfitting in machine learning models.

- **English:**
 - **Overfitting** occurs when a model **learns training data too well**, making it perform poorly on new data.
- **Kinyarwanda:**
 - **Overfitting** iba iyo model **yiga cyane ku makuru yatojwemo**, bigatuma idakora neza ku makuru mashya.

100. What is underfitting, and how does it affect model performance?

- **English:**
 - **Underfitting** happens when a model is **too simple** and fails to capture patterns, leading to **poor accuracy**.
- **Kinyarwanda:**
 - **Underfitting** iba iyo **model itarize bihagije**, bikayituma **itanga ibisubizo bidahuye n'ukuri**.

Unit 1: Apply Data Pre-processing

1.1 Description of Machine Learning Concepts

1. What is machine learning?

- **English:**

Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to learn from data and improve performance on tasks without being explicitly programmed.
- **Kinyarwanda:**

Machine learning (ML) ni ishami rya **artificial intelligence (AI)** rigamiye gufasha mudasobwa kwiga ku makuru no gutera imbere mu mirimo idakeneye amabwiriza yihariye.

2. Define the term "artificial intelligence."

- **English:**

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines, enabling them to perform tasks that typically require human intelligence, such as problem-solving, decision-making, and pattern recognition.
- **Kinyarwanda:**

Artificial Intelligence (AI) bivuze gukoresha mudasobwa n'algorithms kugira ngo zikore ibintu bisaba ubwenge bwa muntu nko gufata imyanzuro, gukemura ibibazo, no gutahura imiterere y'ibintu.

3. How is machine learning related to artificial intelligence?

- **English:**

Machine learning is a subset of artificial intelligence. AI is the broader field that includes rule-based systems and reasoning, while machine learning focuses on algorithms that learn from data to make predictions and decisions.
- **Kinyarwanda:**

Machine Learning ni igice cya AI. AI irimo uburyo bwose bwo gutuma mudasobwa ifata imyanzuro, naho ML yibanda ku kwiga ku makuru kugira ngo itange ibisubizo idahawe amabwiriza yose.

4. List the seven major steps in the machine learning life cycle.

- **English:**
 1. **Data Collection** – Gathering relevant data.
 2. **Data Preprocessing** – Cleaning and transforming data.
 3. **Feature Engineering** – Selecting useful features.
 4. **Model Selection** – Choosing an appropriate algorithm.
 5. **Model Training** – Teaching the model using training data.
 6. **Model Evaluation** – Assessing performance on test data.
 7. **Model Deployment & Maintenance** – Using and updating the model.
- **Kinyarwanda:**
 1. **Gutera amakuru (Data Collection)**
 2. **Gusukura no gutunganya amakuru (Data Preprocessing)**
 3. **Guhitamo ibintu bifasha (Feature Engineering)**
 4. **Guhitamo algorithm nziza (Model Selection)**
 5. **Gutoza mudasobwa (Model Training)**
 6. **Kugenzura uko ikora (Model Evaluation)**
 7. **Gushyira mu bikorwa no kuvugurura (Model Deployment & Maintenance)**

5. What is the purpose of the data gathering phase in the machine learning life cycle?

- **English:**

Data gathering ensures that the machine learning model has enough quality data to learn from and make accurate predictions.
- **Kinyarwanda:**

Gutera amakuru bigamije gutuma mudasobwa igira amakuru ahagije kandi meza kugira ngo yige neza kandi itange ibisubizo bifatika.

6. Explain the difference between data preparation and data wrangling.

- **English:**
 - **Data Preparation** – General cleaning and formatting of data for analysis.
 - **Data Wrangling** – Transforming and restructuring raw data to make it suitable for a model.
- **Kinyarwanda:**
 - **Data Preparation** – Gukora isuku rusange y’amakuru.
 - **Data Wrangling** – Guhindura no kuvugurura amakuru atari mu buryo bwiza kugira ngo akoreshwe na model.

7. What are common issues found in real-world data?

- **English:**

Missing values, duplicates, outliers, inconsistent formats, and biased data.
- **Kinyarwanda:**

Ibibazo rusange ni kubura amakuru, amakuru yikubiyemo, outliers (imibare ibangamye), imiterere idahuje, ndetse n’amakuru arimo ivangura.

8. Why is data cleaning important before model training?

- **English:**
Data cleaning removes errors, fills missing values, and improves model accuracy.
- **Kinyarwanda:**
Isuku y'amakuru ni ingenzi kugira ngo amakosa akurwemo, amakuru abuze yuzuzwe, bityo model igire ubunyamwuga mu bisubizo.

9. Describe two analytical techniques used in data analysis.

- **English:**
 - **Descriptive Analytics** – Summarizing data (e.g., mean, median).
 - **Predictive Analytics** – Using past data to predict future trends.
- **Kinyarwanda:**
 - **Descriptive Analytics** – Gusobanura amakuru (nko kumenya impuzandengo).
 - **Predictive Analytics** – Gukoresha amakuru ya kera kugira ngo hamenyekane ibizaba mu gihe kizaza.

10. What is the goal of the model testing phase in the machine learning life cycle?

- **English:**
To check the model's accuracy and ensure it performs well on unseen data.
- **Kinyarwanda:**
Kugenzura niba model ikora neza kandi ikatanga ibisubizo bifatika ku makuru mashya.

11. Explain what deployment means in the context of machine learning.

- **English:**
Deployment is the process of making the trained ML model available for use in real-world applications.
- **Kinyarwanda:**
Deployment ni ugushyira mu bikorwa model yatojwe ku buryo ishobora gukoreshwa mu buzima busanzwe.

12. Give three real-world applications of machine learning.

- **English:**
 1. Fraud detection in banking
 2. Medical diagnosis
 3. Self-driving cars
- **Kinyarwanda:**
 1. Gutahura uburiganya mu mabanki
 2. Gusuzuma indwara
 3. Imodoka zigenda zidafite abashoferi

13. How is image recognition used in machine learning applications?

- **English:**
Image recognition uses ML models to identify and classify objects, faces, or patterns in images. It is widely used in facial recognition, medical imaging, and security surveillance.
- **Kinyarwanda:**
Kumenya amashusho bikoresha ML kugira ngo imenye ibintu, amasura y'abantu, cyangwa ibindi bimenyetso mu mashusho. Ikoreshwa mu kumenya amasura, mu isuzuma ry'indwara, no mu bikorwa by'umutekano.

14. What is the role of machine learning in virtual personal assistants like Siri or Alexa?

- **English:**
Machine learning enables assistants like Siri and Alexa to understand human language, recognize voice patterns, and improve responses based on user interactions.
- **Kinyarwanda:**
ML ifasha Siri, Alexa, n'izindi gufata no gusobanukirwa n'amagambo y'abantu, gutahura uburyo ijwi riboneka, no gukosora ibisubizo hashingiwe ku mikoreshereze y'abakoresha.

15. Describe one advantage and one disadvantage of using machine learning for fraud detection.

- **English:**
 - **Advantage:** It can detect fraud faster and more accurately than humans.
 - **Disadvantage:** It can sometimes produce false positives, flagging legitimate transactions as fraudulent.
- **Kinyarwanda:**
 - **Ibyiza:** Itahura uburiganya vuba kandi neza kurusha abantu.
 - **Ibibi:** Hari igihe ishobora gutanga ibisubizo bitari byo (false positives) igafata ubucuruzi bwemewe nk'uburiganya.

16. What are some advantages of using machine learning in online shopping?

- **English:**
 - Personalized recommendations
 - Fraud detection
 - Improved customer experience through chatbots
- **Kinyarwanda:**
 - Gutanga inama zijyanye n'icyo umukiriya akunda
 - Gutahura uburiganya mu kwishyura
 - Gufasha abakiriya binyuze muri chatbots

17. Explain the term 'trend analysis' in the context of machine learning.

- **English:**
Trend analysis refers to using historical data to identify patterns and predict future trends, such as in stock prices, sales, or customer behavior.
- **Kinyarwanda:**
Trend analysis isobanura gukoresha amakuru ya kera kugira ngo hamenyekane imiterere y'ibintu no guteganya ibizaba mu gihe kizaza, nko mu biciro by'imigabane, kugurisha cyangwa imico y'abakiriya.

18. What is the difference between machine learning and deep learning?

- **English:**
 - **Machine Learning:** Includes all algorithms that learn from data.
 - **Deep Learning:** A subset of ML that uses neural networks to learn complex patterns.
- **Kinyarwanda:**
 - **Machine Learning:** Ni urwego rusange rw'algorithms zose zigisha ku makuru.
 - **Deep Learning:** Ni igice cya ML gikoresha neural networks kugira ngo kimenye imiterere ikomeye y'amakuru.

19. How does supervised learning differ from unsupervised learning?

- **English:**
 - **Supervised Learning:** Uses labeled data (e.g., classifying spam emails).
 - **Unsupervised Learning:** Finds patterns in unlabeled data (e.g., clustering customers).
- **Kinyarwanda:**
 - **Supervised Learning:** Ikoresha amakuru afite ibisobanuro (nko gutandukanya spam).
 - **Unsupervised Learning:** Igenzura amakuru adafite ibisobanuro igashaka imiterere y'ibintu.

20. Give an example of a machine learning problem best solved with reinforcement learning.

- **English:**
 - Self-driving cars use reinforcement learning to learn navigation through trial and error.
- **Kinyarwanda:**
 - Imodoka zidafite abashoferi zikoresha reinforcement learning kugira ngo zimenye kugenda neza mu mihanda hifashishijwe trial and error.

21. What are the main components of a machine learning algorithm?

- **English:**
 1. **Dataset** – The collection of data.
 2. **Features** – The variables or attributes used for prediction.

3. **Algorithm** – The mathematical model.
4. **Loss Function** – Measures errors in predictions.
5. **Optimizer** – Adjusts model parameters to improve accuracy.
- **Kinyarwanda:**
 1. **Dataset** – Urutonde rw'amakuru.
 2. **Features** – Ibintu bigize amakuru byifashishwa mu iteganyagihe.
 3. **Algorithm** – Modeli ibara ibisubizo.
 4. **Loss Function** – Igipimo cy'amakosa mu bisubizo.
 5. **Optimizer** – Igahindura parameters za model kugira ngo itange ibisubizo byiza.

22. How do machine learning models improve over time?

- **English:**

They learn from new data and adjust predictions based on feedback.
- **Kinyarwanda:**

Ziga ku makuru mashya kandi zikagabanya amakosa hashingiwe ku bisubizo bitangwa.

23. Name two industries where machine learning is commonly applied.

- **English:**
 - Healthcare (disease prediction)
 - Finance (fraud detection)
- **Kinyarwanda:**
 - Ubuvuzi (gutahura indwara)
 - Imari (gutahura uburiganya)

24. What are the key differences between supervised, unsupervised, and semi-supervised learning?

- **English:**
 - **Supervised:** Labeled data
 - **Unsupervised:** No labels
 - **Semi-supervised:** Combination of both
- **Kinyarwanda:**
 - **Supervised:** Amakuru afite ibisobanuro
 - **Unsupervised:** Amakuru adafite ibisobanuro
 - **Semi-supervised:** Ivanzemo ibice byombi

25. Which machine learning type would be best for spam email classification? Why?

- **English:**

Supervised learning is best because we have labeled emails (spam or not).
- **Kinyarwanda:**

Supervised learning niyo ikwiriye kuko dufite e-mails zifite ibisobanuro (spam cyangwa ntibibe spam).

1.2 Preparing Machine Learning Environment

26. Describe the role of reinforcement learning in autonomous driving.

- **English:**
Reinforcement learning helps self-driving cars learn from their environment by rewarding good decisions (e.g., stopping at red lights) and penalizing bad decisions (e.g., running a red light).
- **Kinyarwanda:**
Reinforcement learning ifasha imodoka zitagira abashoferi kwiga uko zigenda mu muhanda: iyo imodoka ikora ibyiza (nko guhagarara ku itara ritukura), irashimirwa, naho iyo ikora amakosa (nko kurenga itara), ihanwa.

27. What is the purpose of model evaluation in the machine learning lifecycle?

- **English:**
Model evaluation checks how well a trained model performs on unseen data, ensuring it is accurate, generalizes well, and is not overfitting.
- **Kinyarwanda:**
Gusuzuma model bigamije kureba niba model yatojwe ikora neza ku makuru mashya, kandi ikagaragaza ubunyamwuga (accuracy) idafite overfitting.

28. List three disadvantages of machine learning.

- **English:**
 1. Data dependency – Models need lots of data.
 2. Computational cost – Requires high processing power.
 3. Bias and fairness issues – Models can inherit bias from training data.
- **Kinyarwanda:**
 1. Ikenera amakuru menshi.
 2. Isaba ubushobozi bwo gucunga mudasobwa zikomeye.
 3. Ishobora kugira ivangura niba amakuru yatojwemo arimo ivangura.

29. Explain how machine learning reduces human error in data analysis.

- **English:**
ML automates data processing, removing human biases and errors, which improves accuracy in analysis.
- **Kinyarwanda:**
Machine learning ituma amakosa akorwa n'abantu agabanuka kuko yikoresha algorithms zidafite amarangamutima, bityo igatanga ibisubizo byizewe.

30. What are common challenges faced when interpreting machine learning results?

- **English:**
 - Lack of explainability (black-box models)
 - Data bias leading to skewed outcomes
 - Overfitting causing poor performance on new data
- **Kinyarwanda:**
 - Kudasobanukirwa neza n'uko models zafashe imyanzuro (black-box)
 - Ivangura mu makuru rishobora gutuma ibisubizo bitaba nyabyo
 - Overfitting ikaba intandaro yo kudatanga ibisubizo byiza ku makuru mashya

1.3 Data Collection and Acquisition

31. What is the first step in preparing a machine learning environment?

- **English:**

Installing a programming language (e.g., Python) and setting up necessary tools (IDE, libraries, frameworks).
- **Kinyarwanda:**

Intambwe ya mbere ni ugushyiraho ururimi rwa porogaramu (nko muri Python) no kwinjiza ibikoresho nk'IDE, libraries, na frameworks.

32. Name three popular Python IDEs used for machine learning.

- **English:**
 1. Jupyter Notebook
 2. PyCharm
 3. VS Code
- **Kinyarwanda:**
 1. Jupyter Notebook
 2. PyCharm
 3. VS Code

33. Describe the purpose of the Anaconda distribution in machine learning.

- **English:**

Anaconda is a Python distribution that comes with pre-installed ML libraries (e.g., NumPy, Pandas, TensorFlow) to simplify the setup of a data science environment.
- **Kinyarwanda:**

Anaconda ni distribution ya Python ifite libraries zikenewe nka NumPy, Pandas, na TensorFlow, byorohereza abashakashatsi gutegura ibidukikije bya data science.

34. What is the role of a package manager like pip in Python?

- **English:**

pip is used to install, update, and manage Python libraries.
- **Kinyarwanda:**

pip ikoreshwa mu gushyiraho, kuvugurura no gucunga libraries za Python.

35. List the basic hardware requirements for installing Python.

- **English:**
At least 4GB RAM, a 64-bit processor, and a compatible operating system (Windows, Mac, or Linux).
- **Kinyarwanda:**
RAM ya 4GB nibura, processor ya 64-bit, hamwe n'uburyo bwo gukoresha nka Windows, Mac, cyangwa Linux.

36. Explain how to verify if Python has been installed successfully on a computer.

- **English:**
Open Command Prompt (or Terminal) and type `python --version` to display the Python version.
- **Kinyarwanda:**
Fungura Command Prompt (cyangwa Terminal) wandike `python --version` kugira ngo ubone version ya Python.

37. What is Jupyter Notebook, and why is it popular for machine learning projects?

- **English:**
Jupyter Notebook is an interactive coding environment that allows users to run code in cells, making it ideal for data analysis and visualization.
- **Kinyarwanda:**
Jupyter Notebook ni ahantu ho kwandika code mu buryo bwigenga aho ushobora kugabanya code mu “cells”, bigafasha cyane mu gusesengura no kwerekana amakuru.

38. How do you install machine learning libraries like NumPy and Pandas using pip?

- **English:**
In the terminal, type:

```
pip install numpy pandas
```

- **Kinyarwanda:**
Muri terminal, wandika:

```
pip install numpy pandas
```

39. What is the purpose of using cloud-based platforms like Google Colab?

- **English:**
Google Colab provides a free cloud environment with pre-installed ML libraries and lets you run Python code without local installation.

- **Kinyarwanda:**
Google Colab itanga uburyo bwo gukorera mu bushorishori (cloud) ifite libraries zashyizwemo, bityo ukaba wakora Python udakeneye kuyishyiraho kuri mudasobwa yawe.

40. Describe the steps to create a virtual environment using Anaconda.

- **English:**
 1. Open Anaconda Prompt.
 2. Create a new environment:

```
conda create --name myenv python=3.8
```
 3. Activate the environment:

```
conda activate myenv
```
 4. Install necessary libraries (e.g.,

```
pip install numpy pandas
```

).
- **Kinyarwanda:**
 1. Fungura Anaconda Prompt.
 2. Kora environment nshya ukoresheje:

```
conda create --name myenv python=3.8
```

3. Emeza iyo environment:

```
conda activate myenv
```

4. Shyiramo libraries zikenewe (nko

```
pip install numpy pandas
```

).

41. Why is it important to install IDEs when preparing for machine learning development?

- **English:**
IDEs (e.g., Jupyter, PyCharm, VS Code) provide tools for writing, debugging, and visualizing code efficiently.
- **Kinyarwanda:**
IDEs zitanga uburyo bworoshye bwo kwandika, kugenzura amakosa, no kwerekana code neza, bigafasha abahanga mu ML.

42. What does CI/CD stand for, and how is it related to machine learning workflows?

- **English:**
CI/CD stands for Continuous Integration and Continuous Deployment. It automates testing, updating, and deploying ML models.

- **Kinyarwanda:**
CI/CD (Continuous Integration and Continuous Deployment) ifasha mu gukosora, kuvugurura no gushyira mu bikorwa models za ML mu buryo bwikora.

43. Explain the difference between TensorFlow and PyTorch.

- **English:**
 - **TensorFlow** (by Google) is widely used for **production models**.
 - **PyTorch** (by Facebook) is preferred in research for its **dynamic computational graph** and ease of debugging.
- **Kinyarwanda:**
 - **TensorFlow** ya Google ikoreshwa cyane mu gushyira mu bikorwa models.
 - **PyTorch** ya Facebook ikunze gukoreshwa mu bushakashatsi kubera uburyo bworoshye bwo gukosora amakosa.

44. What is hyperparameter tuning, and when is it performed in the ML process?

- **English:**
Hyperparameter tuning is optimizing settings like learning rate and batch size to improve model accuracy; it is typically done after initial training and evaluation.
- **Kinyarwanda:**
Hyperparameter tuning ni uguhindura neza imiterere ya model (nko guhitamo learning rate na batch size) kugira ngo ibisubizo bibe byiza; ikorwa nyuma yo gutoza no kugenzura model.

45. Name two cloud services commonly used for machine learning model deployment.

- **English:**
 1. AWS SageMaker
 2. Google AI Platform
- **Kinyarwanda:**
 1. AWS SageMaker
 2. Google AI Platform

1.3 Data Collection and Acquisition

46. Define the term "dataset" in the context of machine learning.

- **English:**
A **dataset** is a structured collection of data points used to train, test, or evaluate a machine learning model.
- **Kinyarwanda:**
Dataset ni urutonde rw'amakuru ateguye neza akoreshwa mu gutoza, kugerageza cyangwa kugenzura model.

47. What are the six V's of Big Data?

- **English:**
 1. **Volume** – Large amounts of data.
 2. **Velocity** – Speed at which data is generated.
 3. **Variety** – Different data types (text, images, etc.).
 4. **Veracity** – Trustworthiness of data.
 5. **Value** – Usefulness of data.
 6. **Variability** – Changing nature of data.
- **Kinyarwanda:**
 1. **Volume** – Amakuru menshi.
 2. **Velocity** – Umuvuduko w’ukuntu amakuru yinjira.
 3. **Variety** – Ubwoko butandukanye bw’amakuru.
 4. **Veracity** – Ukuri cyangwa icyizere cy’amakuru.
 5. **Value** – Akamaro k’amakuru.
 6. **Variability** – Uburyo amakuru ahinduka.

48. Differentiate between structured and unstructured data with examples.

- **English:**
 - **Structured Data:** Organized (rows and columns, e.g., a database table).
 - **Unstructured Data:** Unorganized (e.g., emails, images, videos).
- **Kinyarwanda:**
 - **Structured Data:** Amakuru ateguye neza (nko muri database).
 - **Unstructured Data:** Amakuru adateguwe (nko emails, amashusho, videwo).

49. Explain the importance of data acquisition in machine learning.

- **English:**

Data acquisition ensures that models have high-quality and relevant data, which is crucial for accurate predictions.
- **Kinyarwanda:**

Gutoranya amakuru bifasha models kugira amakuru meza kandi ajyanye n’icyo bakora, bikaba ingenzi mu gutanga ibisubizo byizewe.

50. What is the role of IoT sensors in data collection for ML models?

- **English:**

IoT sensors automatically collect real-time data (e.g., temperature, motion) which can be used to train ML models.
- **Kinyarwanda:**

IoT sensors zitanga amakuru mu buryo bwikora kandi mu gihe nyacyo (nko kumenya ubushyuhwe cyangwa imikorere y’ibikoreshe) bikoreshe mu gutoza models.

1.4 Interpret Data Visualization

51. Describe how smartphones can be used as a data source in machine learning.

- **English:**
Smartphones collect data via sensors (GPS, accelerometers, microphones) and user interactions (app usage, typing patterns) for ML applications like health monitoring and personalized recommendations.
- **Kinyarwanda:**
Telefoni zigendanwa zitanga amakuru zifashishije sensors (GPS, accelerometers, microfone) ndetse n'imikorere y'abakoresha (gukoresha applications, uburyo bandika) ikoreshwa mu mishinga ya ML nko kugenzura ubuzima no gutanga inama z'icyo umukiriya akunda.

52. What is a data warehouse, and how is it different from a traditional database?

- **English:**
 - **Data Warehouse:** Stores historical and aggregated data for analysis.
 - **Traditional Database:** Stores real-time transactional data for day-to-day operations.
- **Kinyarwanda:**
 - **Data Warehouse:** Ikusanya amakuru ya kera hamwe n'ayahurijwe hamwe hagamijwe gusesengura.
 - **Database isanzwe:** Ibika amakuru y'ibikorwa bya buri muni.

53. List three challenges faced during the data collection phase.

- **English:**
 1. Incomplete data (**missing values**)
 2. Data privacy issues (**legal restrictions**)
 3. Inconsistent data formats
- **Kinyarwanda:**
 1. Amakuru atuzuye
 2. Ibibazo by'amategeko byo gutoranya amakuru
 3. Imiterere itajyanye y'amakuru

54. What is the difference between primary data and secondary data sources?

- **English:**
 - **Primary Data:** Collected directly from original sources (e.g., surveys).
 - **Secondary Data:** Data that has already been collected and published (e.g., reports).
- **Kinyarwanda:**
 - **Primary Data:** Amakuru yatoranijwe bwa mbere (nko gukora ubushakashatsi).
 - **Secondary Data:** Amakuru yakusanyijwe mbere akaba yaratangajwe (nko mu byegeranyo).

55. Explain the term "data labeling" in supervised machine learning.

- **English:**
Data labeling is the process of assigning correct categories or tags to data so that a model can learn patterns accurately.
- **Kinyarwanda:**
Data labeling ni uguhereza amakuru ibisobanuro byayo (tag cyangwa category) kugira ngo model ibashe kwiga imiterere y'amakuru.

56. What are transactional data, and how are they relevant in machine learning?

- **English:**
Transactional data are records of business transactions (e.g., bank transactions, online purchases) and are used in fraud detection and customer behavior analysis.
- **Kinyarwanda:**
Transactional data ni inyandiko zerekana ibikorwa by'ubucuruzi (nko mu mabanki cyangwa kugura kuri internet) zikaba zikoreshwa mu gutahura uburiganya no gusesengura imico y'abakiriya.

57. Describe the characteristics of Big Data.

- **English:**
Big Data is characterized by the 6 V's: **Volume, Velocity, Variety, Veracity, Value,** and **Variability.**
- **Kinyarwanda:**
Big Data ifite ibiranga 6 V's:
 1. **Volume** – Amakuru menshi
 2. **Velocity** – Umuvuduko w'amakuru
 3. **Variety** – Ubwoko butandukanye
 4. **Veracity** – Ukuri kw'amakuru
 5. **Value** – Ibyo amakuru amariye
 6. **Variability** – Uburyo amakuru ahinduka

58. How can APIs be used to collect data for machine learning projects?

- **English:**
APIs allow **fetching real-time data** from **external sources** (websites, databases, IoT devices) for use in ML projects.
- **Kinyarwanda:**
APIs zituma haboneka amakuru y'igihe nyacyo ava hanze (ku mbuga, muri databases, cyangwa ku bikoresho bya IoT) bikoreshwa mu mishinga ya ML

59. What is data augmentation, and when is it applied in ML projects?

- **English:**
Data augmentation is the process of artificially increasing the size of a dataset by applying transformations (e.g., rotating images) to improve model generalization.

- **Kinyarwanda:**
Data augmentation ni ugukora impinduka ku makuru (nko guhindura amashusho) kugira ngo dataset irusheho kuba nini kandi model ibe ishoboye kwiga neza.

60. Describe the purpose of using platforms like Kaggle for data acquisition.

- **English:**
Kaggle is an online platform providing datasets, competitions, and collaborative projects that help researchers find and use data for ML.
- **Kinyarwanda:**
Kaggle ni urubuga rutanga datasets, amarushanwa, n'imishinga y'ubufatanye ifasha abashakashatsi kubona no gukoresha amakuru mu mishinga ya ML.

1.4 Interpret Data Visualization (Continued)

61. What is data visualization, and why is it important in machine learning?

- **English:**
Data visualization is the process of graphically representing data (using charts, graphs, and plots) to identify patterns and insights.
- **Kinyarwanda:**
Data visualization ni ugushyira amakuru mu buryo bw'ishusho (amaseti, graphs, plots) kugira ngo hamenyekane imiterere n'ibitekerezo mu makuru.

62. Name three popular data visualization libraries in Python.

- **English:**
 1. Matplotlib
 2. Seaborn
 3. Plotly
- **Kinyarwanda:**
 1. Matplotlib
 2. Seaborn
 3. Plotly

63. What is Matplotlib, and how is it used in data visualization?

- **English:**
Matplotlib is a Python library used for creating static, animated, and interactive visualizations.
- **Kinyarwanda:**
Matplotlib ni library ya Python ikoreshwa mu gukora amashusho y'amakuru (static, animated, interactive).

64. Explain the difference between Matplotlib and Seaborn.

- **English:**
 - Matplotlib provides basic plotting functions.
 - Seaborn, built on Matplotlib, offers more advanced and visually appealing charts.
- **Kinyarwanda:**
 - Matplotlib itanga uburyo bw'ibanze bwo gushushanya.
 - Seaborn yubatswe kuri Matplotlib, itanga amashusho arambuye kandi ateye amabengeza.

65. How does Plotly differ from traditional static visualization tools?

- **English:**

Plotly creates interactive visualizations that allow users to zoom, hover, and dynamically explore data.
- **Kinyarwanda:**

Plotly ikora amashusho y'amakuru afite ubushobozi bwo guhinduranya (interactive), aho ushobora kugenda mu ishusho no kugenzura amakuru mu buryo bw'igihe nyacyo.

66. What type of data is best visualized using a scatter plot?

- **English:**

A **scatter plot** is best for displaying relationships between two numerical variables.
- **Kinyarwanda:**

Scatter plot ikoresheya mu kugaragaza isano hagati y'imibare ibiri.

67. When would you use a heatmap in data analysis?

- **English:**

Heatmaps are used to show the intensity of values and correlations between variables.
- **Kinyarwanda:**

Heatmap ikoresheya mu kugaragaza uko agaciro kagenda kigaragara n'uburyo ibintu bihurira hamwe.

68. Describe the purpose of using dashboards like Tableau or Power BI.

- **English:**

Dashboards provide interactive, real-time visual summaries of data to monitor business performance or ML model outputs.
- **Kinyarwanda:**

Dashboards nka Tableau cyangwa Power BI zitanga ishusho y'amakuru mu buryo bw'igihe nyacyo bikoreshwa mu kugenzura imikorere y'ubucuruzi cyangwa ibisubizo bya models.

69. What is the role of data visualization in identifying data anomalies?

- **English:**
Visualization can quickly reveal outliers, **missing data**, or unusual patterns that indicate anomalies.
- **Kinyarwanda:**
Gushushanya amakuru bifasha gutahura vuba ibitagenda neza nko kureba outliers, amakuru abuze, cyangwa ibindi bimenyetso bitangaje.

70. Explain how a histogram helps in understanding data distribution.

- **English:**
Histograms show the frequency distribution of data, making it easier to see its spread and skewness.
- **Kinyarwanda:**
Histogram yerekana uko imibare igabanijemo, bityo bigafasha kumva niba amakuru ari evenly distributed cyangwa afite imiterere runaka.

71. What are box plots used for in data visualization?

- **English:**
Box plots display the distribution of data based on quartiles and help in identifying outliers.
- **Kinyarwanda:**
Box plots zerekana uko imibare igabanijemo (quartiles) kandi zifasha gutahura outliers.

72. Describe the process of creating a bar chart using Matplotlib.

- **English:**
Example code:

```
import matplotlib.pyplot as plt
categories = ["A", "B", "C"]
values = [10, 20, 15]
plt.bar(categories, values)
plt.show()
```

- **Kinyarwanda:**
Urugero rwa bar chart muri Matplotlib:

```
import matplotlib.pyplot as plt
categories = ["A", "B", "C"]
values = [10, 20, 15]
plt.bar(categories, values)
plt.show()
```

73. What is the importance of labeling axes in data visualization?

- **English:**
Labels help users understand what each axis represents, making the graph more informative.
- **Kinyarwanda:**
Labeling y'inyuguti (axes) ifasha umuntu kumva icyo buri murongo cyangwa umushinga werekana.

74. How can visualizations help in feature selection for machine learning models?

- **English:**
Visualizations help reveal which features have stronger correlations with the target variable, guiding feature selection.
- **Kinyarwanda:**
Amashusho y'amakuru afasha kumenya ibifite agaciro mu kugena ibintu, bityo bigafasha mu guhitamo features nziza.

75. Explain the concept of correlation matrices in data visualization.

- **English:**
A correlation matrix is a table showing correlation coefficients between variables, useful in identifying relationships between features.
- **Kinyarwanda:**
Correlation matrix ni urutonde rwerekana coefficient z'ukuntu ibintu bihuriye, bigafasha kumenya imibanire hagati ya features.

1.5 Perform Data Cleaning

76. What is data cleaning in machine learning?

- **English:**
Data cleaning is the process of removing errors, handling missing values, and fixing inconsistencies in data to improve model performance.
- **Kinyarwanda:**
Data cleaning ni uburyo bwo gukosora amakosa, kuzuza amakuru abuze, no gukemura imiterere idahwitse mu makuru kugira ngo model ikore neza.

77. List three common data quality issues addressed during data cleaning.

- **English:**
 1. Missing values
 2. Duplicate data
 3. Outliers
- **Kinyarwanda:**
 1. Amakuru abuze

2. Amakuru yikubiyemo
3. Outliers

78. Explain the term "missing data" and how it can affect machine learning models.

- **English:**
Missing data refers to incomplete records; it can cause biased predictions and reduced model accuracy.
- **Kinyarwanda:**
Missing data ni amakuru atuzuye, ashobora gutera ibisubizo bitari byo ndetse n'uko model itatanga ibisubizo byizewe.

79. What is data imputation, and when is it necessary?

- **English:**
Data imputation is replacing missing values with estimated ones (like mean or median) and is necessary when data is incomplete yet still valuable.
- **Kinyarwanda:**
Data imputation ni ugusimbuza amakuru abuze n'agaciro kagenewe (nko impuzandengo cyangwa median) kandi ikenewe iyo amakuru adafite ibice byose ariko agifite agaciro.

80. Describe two methods for handling missing values in datasets.

- **English:**
 1. Remove rows/columns with too many missing values.
 2. Impute missing values using statistical methods (mean, median, mode).
- **Kinyarwanda:**
 1. Gukuraho imirongo cyangwa inkingi ifite amakuru make cyane.
 2. Kuzuza amakuru abuze ukoresheje uburyo bwa statistiki (impuzandengo, median, cyangwa mode).

81. What is the difference between duplicate data and inconsistent data?

- **English:**
 - Duplicate data: The same record appears more than once.
 - Inconsistent data: Data that is formatted or recorded differently even though it represents the same information.
- **Kinyarwanda:**
 - Duplicate data: Iyo record imwe yagaragara incuro nyinshi.
 - Inconsistent data: Amakuru agaragara mu buryo butandukanye nubwo ahagarariye ibintu bimwe.

82. How can outliers negatively impact machine learning models?

- **English:**
Outliers can skew the model's understanding, reducing accuracy and leading to poor generalization.
- **Kinyarwanda:**
Outliers zishobora gutuma model itabona neza uko amakuru agomba kwiyitwararika, bigatuma idatanga ibisubizo byizewe.

83. What techniques are used to detect outliers in data?

- **English:**
 1. Box Plots
 2. Z-Score calculation
 3. Interquartile Range (IQR)
- **Kinyarwanda:**
 1. Box Plots
 2. Z-Score
 3. IQR

84. Explain the importance of data normalization in machine learning.

- **English:**
Normalization scales numerical data to a common range, preventing features with large values from dominating the model.
- **Kinyarwanda:**
Normalization ishyira imibare ku gipimo gihwanye, bikarinda ko features zifite agaciro kanini zihiga izindi.

85. What is data transformation, and why is it done during data cleaning?

- **English:**
Data transformation is converting raw data into a format more suitable for modeling, often improving performance.
- **Kinyarwanda:**
Data transformation ni uguhindura amakuru atarimo gutunganywa neza mu buryo bworoshye gukoreshwa na model, bityo bigatuma ibisubizo biba byiza.

86. How do you handle categorical data in machine learning preprocessing?

- **English:**
Categorical data is handled by converting it into numerical form using methods such as Label Encoding or One-Hot Encoding.
- **Kinyarwanda:**
Amakuru afite ibyiciro ahindurwamo imibare ukoresheje uburyo nka Label Encoding cyangwa One-Hot Encoding.

87. Describe the process of feature scaling and why it's important.

- **English:**
Feature scaling adjusts the range of numerical features so they contribute equally to the model, preventing bias towards features with larger scales.
- **Kinyarwanda:**
Feature scaling ni uguhindura imibare kugira ngo yose ibe ku gipimo kimwe, bityo nta feature ifite agaciro kanini ikarenza izindi.

88. What are the potential risks of over-cleaning a dataset?

- **English:**
Over-cleaning can remove valuable information and lead to loss of important patterns in the data.
- **Kinyarwanda:**
Over-cleaning ishobora gukuraho amakuru akenewe, bigatuma habaho gutakaza imiterere y'ingenzi mu makuru.

89. Explain the concept of data leakage and how to prevent it.

- **English:**
Data leakage occurs when information from the test set inadvertently becomes available during training, leading to overly optimistic performance. Prevent it by strictly separating training and test data.
- **Kinyarwanda:**
Data leakage iba iyo amakuru y'igerageza (test) yinjira mu myitozo (training) bitateganijwe, bikabyara ibisubizo by'umucyo. Ibyirinde ukoresheje gutandukanya neza amakuru y'imyitozo n'ay'igerageza.

90. What is underfitting, and how does it affect model performance?

- **English:**
Underfitting happens when a model is **too simple** to capture the underlying patterns, resulting in **poor accuracy**.
- **Kinyarwanda:**
Underfitting iba iyo model itarize bihagije, ikabura gutahura imiterere nyayo y'amakuru, bityo igatanga ibisubizo bidahuye n'ukuri.

Unit 2: Develop Machine Learning Model

2.1 Description of Machine Learning Algorithms and Applications

91. What is a machine learning algorithm?

- **English:**
A machine learning algorithm is a mathematical procedure that learns patterns from data to make predictions or decisions.

- **Kinyarwanda:**
Machine learning algorithm ni uburyo bwa siyansi bwo kwiga ku makuru no gufata imyanzuro.

92. List three types of machine learning algorithms.

- **English:**
 1. Supervised Learning (e.g., Decision Trees, SVM)
 2. Unsupervised Learning (e.g., K-Means, PCA)
 3. Reinforcement Learning (e.g., Q-Learning)
- **Kinyarwanda:**
 1. Supervised Learning (nko Decision Trees, SVM)
 2. Unsupervised Learning (nko K-Means, PCA)
 3. Reinforcement Learning (nko Q-Learning)

93. Explain the difference between classification and regression algorithms.

- **English:**
 - **Classification predicts** categories (e.g., spam vs. non-spam).
 - **Regression predicts continuous** values (e.g., house prices).
- **Kinyarwanda:**
 - **Classification** igena ibyiciro (spam cyangwa ntibibe spam).
 - **Regression** itanga imibare ihoraho (nko kumenya igiciro cy'inzu).

94. What is the main objective of clustering algorithms?

- **English:**
Clustering algorithms group similar data points together without predefined labels.
- **Kinyarwanda:**
Clustering algorithms itandukanya amakuru ashingiye ku isano adafite labels zateganijwe.

95. Describe how decision trees work in machine learning.

- **English:**
Decision trees split data into **smaller groups** based on decision rules until a final prediction or classification is reached.
- **Kinyarwanda:**
Decision trees zigabanya amakuru mu matsinda mato ukoresheje amabwiriza kugeza igihe hagezwe ku cyiciro cyangwa ku iteganyagihe.

96. What are the advantages of using Random Forest over a single decision tree?

- **English:**
Random Forest uses multiple **decision trees** to improve accuracy and **reduce overfitting**.

- **Kinyarwanda:**
Random Forest ihuza decision trees nyinshi, bigatuma ibisubizo biba byiza kandi bigatuma **hatabaho overfitting**.

97. Define the term "support vector machine" (SVM).

- **English:**
SVM is an algorithm that finds the **optimal hyperplane separating** two classes in the feature space.
- **Kinyarwanda:**
SVM ni algorithm ifasha gushaka umurongo cyangwa **hyperplane** utandukanya ibyiciro bibiri mu makuru.

98. In which scenario would you prefer to use logistic regression over linear regression?

- **English:**
Logistic regression is used for **classification problems** (predicting discrete outcomes), while **linear regression** is used for **continuous predictions**.
- **Kinyarwanda:**
Logistic regression ikoresheya mu byiciro (**classification**), naho **linear regression** ikoresheya mu mibare ihoraho (**continuous values**).

99. What is the purpose of using K-Nearest Neighbors (KNN) in machine learning?

- **English:**
KNN classifies a data point based on the majority class among its nearest neighbors.
- **Kinyarwanda:**
KNN igenzura icyiciro cya data hashingiye ku byiciro byinshi biri hafi yayo.

100. Explain the concept of overfitting in machine learning models.

- **English:**
Overfitting occurs when a model learns the **training data too well** including noise so it performs **poorly on new, unseen data**.
- **Kinyarwanda:**
Overfitting iba iyo model yiga amakuru yatojwemo ku buryo buhambaye (harimo n'amakosa) bityo ntikore neza ku makuru mashya.

Unit 2.2: Selection of Machine Learning Algorithms

101. What factors should be considered when selecting a machine learning algorithm?

- **English:**
Factors include the **size of the dataset**, **types of features**, **interpretability requirements**, **training time**, **accuracy**, **model complexity**, and **scalability**.
- **Kinyarwanda:**
Ibintu by'ingenzi ni ingano ya dataset, ubwoko bwa features, uburyo bwo gusobanura ibisubizo, igihe cyo gutoza, ukuri, imiterere y'algorithms, ndetse n'ubushobozi bwo gukwirakwiza.

102. How does the size of a dataset influence algorithm selection?

- **English:**
Large datasets can support **complex, high-variance models**, whereas **small datasets** may require **simpler models** to **avoid overfitting**.
- **Kinyarwanda:**
Dataset nini ishobora gutuma hakoreshwa algorithms zikomeye kandi zifite variance nyinshi, mu gihe **dataset ntoya** zikenere models zoroshye kugira ngo hirindwe overfitting.

103. What type of algorithm is best suited for high-dimensional data?

- **English:**
Algorithms such as **Support Vector Machines** (with appropriate kernels) or those preceded by dimensionality reduction (like PCA) are suitable for **high-dimensional data**.
- **Kinyarwanda:**
Algorithms nka **SVM** (ifite kernel ikwiriye) cyangwa uburyo bwo kugabanya dimensions (nko gukoresha PCA) biba bikwiriye ku makuru afite dimensions nyinshi.

104. Explain the bias-variance trade-off in machine learning.

- **English:**
Bias is the error from overly **simple assumptions**, while **variance** is the **error due** to model **sensitivity to fluctuations**.

Balancing them is essential for good generalization.

- **Kinyarwanda:**
Bias ni amakosa aturuka ku gusobanura ibintu byoroshye cyane, naho **variance** ni amakosa aturuka ku guhinduka kwayo. Kugabanya ayo makosa byombi ni ingenzi kugira ngo model isobanukire neza.

105. How does model complexity affect bias and variance?

- **English:**
More complex models typically have low bias (**fit training data well**) but **high variance** (**sensitive to noise**), whereas

simpler models have higher bias but **lower variance**.

- **Kinyarwanda:**
Models zikomeye ziba zifite bias nke ariko variance nyinshi, naho **models zoroshye** zikagira bias nyinshi ariko variance nke.

106. What is the difference between parametric and non-parametric models?

- **English:**
Parametric models have a fixed number of parameters regardless of data size, while non-parametric models can grow in complexity as more data is provided.
- **Kinyarwanda:**
Parametric models zifite umubare w'imibare ugenwe, naho non-parametric models ziyongera mu miterere bitewe n'ubwinshi bw'amakuru.

107. When would you choose a non-parametric algorithm like KNN?

- **English:**
When the data does not follow a known distribution or the relationship is complex, **non-parametric** models like KNN offer flexibility without assuming a fixed form.
- **Kinyarwanda:**
Iyo amakuru atakurikiranye uburyo buzwi cyangwa imiterere ari imburagihe, gukoresha algorithms nka KNN bitanga ubwisanzure budakeneye guhindura imiterere ihamye.

108. Explain how feature selection impacts algorithm performance.

- **English:**
Feature selection reduces noise, lowers computational cost, and prevents overfitting by focusing on the most informative attributes.
- **Kinyarwanda:**
Guhitamo features neza bigabanya urusaku, bigatuma computation igenda neza, kandi bigafasha kwirinda overfitting kuko bigena iby'ingenzi gusa.

109. What is the curse of dimensionality, and how can it affect machine learning models?

- **English:**
The curse of dimensionality refers to the problems that arise when data has too many features, such as increased sparsity and computational burden, which may lead to overfitting.
- **Kinyarwanda:**
Curse of dimensionality ni ikibazo kivuka iyo amakuru afite features nyinshi, bigatera kubura umubare w'amakuru (sparsity) no kongera imbaraga zikoreshwa, bishobora gutera overfitting.

110. How do cross-validation techniques help in selecting the best algorithm?

- **English:**
Cross-validation splits the data into several parts, training and testing repeatedly to provide a robust estimate of model performance and reduce overfitting risk.
- **Kinyarwanda:**
Cross-validation igabanya amakuru mu bice byinshi, igakoresha buri gice nk'icy'imitozo n'igerageza mu buryo buhoraho, bigatanga igipimo cy'ukuri cya model kandi bigafasha kwirinda overfitting.

111. What is k-fold cross-validation, and how is it implemented?

- **English:**
k-Fold cross-validation divides the dataset into k equal folds; each fold is used as the test set once while the remaining folds train the model.
- **Kinyarwanda:**
k-Fold cross-validation igabanya dataset mu bice k bingana; buri gice kigakoreshwa nk'igerageza rimwe naho ibisigaye bigakoreshwa mu myitozo, bikazungurana inshuro k.

112. Describe the leave-one-out cross-validation method.

- **English:**
Leave-one-out cross-validation (LOOCV) uses every single data point as a test case one at a time, with all remaining points used for training, and repeats this for all data points.
- **Kinyarwanda:**
Leave-one-out cross-validation (LOOCV) ikoreshwa aho buri kantu k'amakuru kagira umwanya w'igerageza, ibisigaye bigakoreshwa mu myitozo, bikaba bigakorwa kuri buri kantu.

113. How do you evaluate the performance of a classification algorithm?

- **English:**
Performance is evaluated using **metrics** such as **accuracy, precision, recall, F1-score**, and by **examining the confusion matrix**.
- **Kinyarwanda:**
Uko model yo mu byiciro ikora bipimwa ukoresheje metrics nka accuracy, precision, recall, F1-score, ndetse na confusion matrix.

114. What is the confusion matrix, and how is it interpreted?

- **English:**
A confusion matrix is a table that shows the number of correct and incorrect predictions segmented by true positives, true negatives, false positives, and false negatives.
- **Kinyarwanda:**
Confusion matrix ni urutonde rwerekana uko model yakoze mu gutahura; igaragaza true positives, true negatives, false positives, na false negatives.

115. Define precision, recall, and F1-score in classification models.

- **English:**
 - **Precision:** $\frac{TP}{(TP+FP)}$ – the fraction of positive predictions that are correct.
 - **Recall:** $\frac{TP}{(TP+FN)}$ – the fraction of actual positives correctly predicted.
 - **F1-score:** The harmonic mean of precision and recall.
- **Kinyarwanda:**
 - **Precision:** Igipimo cy'ukuri mu byavuzwe ko ari positives, kikaba $\frac{TP}{(TP+FP)}$.
 - **Recall:** Igipimo cy'ukuri mu by'ukuri positives, kikaba $\frac{TP}{(TP+FN)}$.
 - **F1-score:** Impuzandengo ya harmonic hagati ya precision na recall.

116. When would you prioritize precision over recall in a model?

- **English:**

When **the cost of false positives is high** (e.g., spam detection or medical diagnoses where a false alarm is expensive).
- **Kinyarwanda:**

Iyo amakosa yo kuvuga ibintu bidahuye (false positives) afite igiciro kinini, nko mu gutahura spam cyangwa mu buvuzi aho amakosa ashobora kugira ingaruka zikomeye.

117. Explain the ROC curve and its significance in model evaluation.

- **English:**

The ROC (Receiver Operating Characteristic) curve plots the true positive rate against the false positive rate at various threshold settings, illustrating the trade-off between sensitivity and specificity.
- **Kinyarwanda:**

ROC curve (Receiver Operating Characteristic) yerekana uko true positive rate ihagaze ugereranyije na false positive rate mu bipimo bitandukanye, bigaragaza uburyo sensitivity na specificity bihurira.

118. What is AUC (Area Under the Curve), and how is it interpreted?

- **English:**

AUC measures the total area under the ROC curve; a higher AUC indicates that the model better distinguishes between classes.
- **Kinyarwanda:**

AUC ni igipimo cy'umwanya uri muni ya ROC curve; agaciro kanini kerekana ko model ishobora gutandukanya ibyiciro neza.

119. Describe the role of metrics like mean squared error (MSE) in regression analysis.

- **English:**
MSE calculates the average squared difference between **predicted** and **actual values**, serving as an indicator of a **regression model's accuracy**.
- **Kinyarwanda:**
MSE ipima impuzandengo y'itandukaniro rya kabiri hagati y'ibyo model ivuga n'iby'ukuri, ikaba igipimo cy'uko regression ikora neza.

120. How does R-squared (R^2) measure the performance of a regression model?

- **English:**
R-squared indicates the proportion of the variance in the dependent variable that is explained by the independent variables; a value close to 1 implies a good fit.
- **Kinyarwanda:**
R-squared (R^2) igaragaza igipimo cy'uko impinduka mu kintu cyifuzwa zisobanurwa n'ibindi bintu; agaciro kari hafi ya 1 kerekana ko model ikora neza.

Unit 2.3: Train Machine Learning Model

121. What is the purpose of training a machine learning model?

- **English:**
Training teaches the model to learn patterns from data so it can make accurate predictions on new, unseen data.
- **Kinyarwanda:**
Gutoza model bigamije kuyigisha kumenya imiterere y'amakuru kugira ngo itange ibisubizo byizewe ku makuru mashya.

122. Explain the difference between training data and testing data.

- **English:**
Training data is used to teach the model, whereas **testing data** is used to evaluate its performance on unseen data.
- **Kinyarwanda:**
Amakuru y'imyitoto akoreshwa mu kwigisha model, naho **amakuru y'igerageza** akoreshwa mu gupima uko ikora ku makuru mashya.

123. What is model training, and how does it differ from model evaluation?

- **English:**
Model training adjusts the model's internal parameters using the training data, while model evaluation measures its performance using separate testing or validation data.
- **Kinyarwanda:**
Gutoza model ni uguhindura parameters zayo ukoresheje amakuru y'imyitoto, naho kugenzura model ni ugupima uko ikora ukoresheje amakuru y'igerageza cyangwa validation.

124. Describe the process of splitting a dataset into training and testing sets.

- **English:**
The **dataset is partitioned** commonly using ratios such as **70–80% for training** and **20–30% for testing** to ensure that the model is evaluated on unseen data.
- **Kinyarwanda:**
Dataset igabanywa mu bice, akenshi hakoreshwa urugero rwa **70–80%** ku myitozo na 20–30% ku igerageza, kugira ngo model ipimwe ku makuru atigeze akoreshwa.

125. Why is it important to avoid data leakage during model training?

- **English:**
Data leakage allows the model to access information from the test set during training, leading to overly optimistic performance and poor generalization.
- **Kinyarwanda:**
Data leakage ituma model ibona amakuru y'igerageza mu myitozo, bigatera ibisubizo by'igishuko (overly optimistic) kandi bigatuma itabasha gukorera neza ku makuru mashya.

126. What is feature scaling, and why is it important before model training?

- **English:**
Feature scaling standardizes the range of independent variables, ensuring that no single feature dominates the learning process.
- **Kinyarwanda:**
Feature scaling ihindura imibare y'ibintu bitandukanye ku gipimo kimwe, bigatuma nta feature imwe igira uruhare runini mu myitozo.

127. Explain the difference between normalization and standardization.

- **English:**
Normalization scales data to a specific range (e.g., 0–1), while standardization transforms data to have a mean of 0 and a standard deviation of 1.
- **Kinyarwanda:**
Normalization ishyira amakuru ku gipimo runaka (nko 0–1), naho standardization ihindura amakuru kugira ngo agire impuzandengo ya 0 n'ukwiyongera kwa 1.

128. What is the role of data augmentation in machine learning?

- **English:**
Data augmentation artificially expands the training dataset by applying transformations, thus improving model generalization.
- **Kinyarwanda:**
Data augmentation yongera ubunini bwa dataset y'imyitozo ikoresheje impinduka zitandukanye, bigatuma model ibasha kwiga neza.

129. How does cross-validation improve model performance during training?

- **English:**
Cross-validation tests the model on multiple data splits, ensuring that the model generalizes well and reducing the risk of overfitting.
- **Kinyarwanda:**
Cross-validation ikoresha mu kugabanya amakuru mu bice bitandukanye, bigatuma model ipimwa neza kandi ikirinda overfitting.

130. Describe the process of gradient descent in model training.

- **English:**
Gradient descent is an iterative optimization algorithm that updates model parameters in the direction that minimizes the loss function.
- **Kinyarwanda:**
Gradient descent ni uburyo bwo guhindura parameters za model mu ntambwe ntoya buhoraho bugamije kugabanya loss function.

131. What is a loss function, and how is it used in model optimization?

- **English:**
A loss function quantifies the difference between predicted and actual values; optimization algorithms minimize this loss during training.
- **Kinyarwanda:**
Loss function ipima itandukaniro hagati y'ibivugwa na byo n'iby'ukuri; algorithms zo mu myitozo zigamije kugabanya iri tandukaniro.

132. Give an example of a common loss function used in regression models.

- **English:**
Mean Squared Error (MSE) is commonly used to measure the average squared difference between predictions and true values.
- **Kinyarwanda:**
Mean Squared Error (MSE) ni urugero rwiza rwo gupima impuzandengo y'itandukaniro rya kabiri hagati y'ibyo model ivuga n'iby'ukuri.

133. How does early stopping prevent overfitting during model training?

- **English:**
Early stopping monitors performance on a validation set and **stops training** when **improvements cease**, thus **avoiding overfitting** to the training data.
- **Kinyarwanda:**
Early stopping ikurikirana uko model ikora ku validation set, igahagarika imyitozo igihe performance itakomeza kunozwa, bityo ikirinda overfitting.

134. What are the benefits of using transfer learning in model development?

- **English:**
Transfer learning leverages pre-trained models on similar tasks, **reducing training** time and often improving performance on related tasks.
- **Kinyarwanda:**
Transfer learning ifasha gukoresha models zatojwe ku mirimo isa, bigatuma igihe cyo gutoza kigabanuka kandi ibisubizo bikaba byiza.

135. Explain how batch size impacts the training of a neural network.

- **English:**
Batch size determines how many samples are processed before updating model parameters; larger batches offer smoother updates, while **smaller batches** provide more frequent but noisier updates.
- **Kinyarwanda:**
Batch size igena umubare w'ibyitegererezo bikorehwa mbere yo kuvugurura parameters; **batch nini** itanga guhinduka kudahindagurika, naho batch ntoya ikatanga updates kenshi ariko zishobora kuba zifite urusaku.

136. What is the role of epochs in training deep learning models?

- **English:**
An epoch is one complete pass through the entire training dataset; **multiple epochs** help the model progressively learn better.
- **Kinyarwanda:**
Epoch ni uko model yiga dataset yose rimwe; epochs nyinshi zifasha model kwiga neza uko imyitozo igenda.

137. Describe the difference between batch gradient descent and mini-batch gradient descent.

- **English:**
Batch gradient descent uses the entire dataset to perform one update per iteration, while **mini-batch gradient descent** uses smaller subsets of data, balancing computational efficiency and stability.
- **Kinyarwanda:**
Batch gradient descent ikoresha dataset yose mu gihe kimwe cyo kuvugurura, naho **mini-batch gradient descent** ikoresha utubice duto, bigahuza umuvuduko no gukosora neza.

138. What is model convergence, and how do you know when a model has converged?

- **English:**
Model convergence is reached when further training yields minimal change in the loss function; the model parameters stabilize.

- **Kinyarwanda:**
Model convergence iba iyo imyitoto itakomeza guhindura loss function cyane, kandi parameters zihagarara.

139. How can you handle imbalanced datasets during model training?

- **English:**
Techniques such as oversampling, undersampling, and using weighted loss functions can address imbalances in the data.
- **Kinyarwanda:**
Ushobora gukoresha oversampling, undersampling, cyangwa weighted loss functions kugira ngo ukemure ikibazo cy'amakuru adahwanye neza.

140. What are synthetic data generation techniques, and when are they useful?

- **English:**
Synthetic data generation creates artificial data samples (using methods like SMOTE) to augment scarce or imbalanced datasets, improving training.
- **Kinyarwanda:**
Synthetic data generation ni uburyo bwo gukora amakuru y'ubwoko bw'ubukorano (nko gukoresha SMOTE) kugira ngo wongere dataset idahagije cyangwa idahuye neza, bityo bigafasha imyitoto.

Unit 2.4: Evaluating Machine Learning Models

141. Why is model evaluation critical in the machine learning lifecycle?

- **English:**
It ensures that the model performs well on unseen data and can generalize, which is crucial for real-world applications.
- **Kinyarwanda:**
Kugenzura model ni ingenzi kuko bituma hamenyekana niba model ikora neza ku makuru mashya kandi ishobora gusobanukirwa neza, bityo ikaba ishobora gukoreshwa mu buzima busanzwe.

142. What is the difference between training accuracy and testing accuracy?

- **English:**
Training accuracy measures performance on the data used to train the model, while testing accuracy measures performance on independent, unseen data.
- **Kinyarwanda:**
Training accuracy ipima uko model ikora ku makuru y'imyitoto, naho testing accuracy ipima uko ikora ku makuru atigeze akoreshwa mu myitoto.

143. How does cross-validation help in model evaluation?

- **English:**
Cross-validation averages performance over multiple data splits, providing a more reliable estimate of model generalization.
- **Kinyarwanda:**
Cross-validation ifata impuzandengo y'uko model ikora mu bice bitandukanye by'amakuru, bigatanga igipimo cy'ukuri cy'uko ikora ku makuru yose.

144. Explain the difference between holdout validation and k-fold cross-validation.

- **English:**
Holdout validation splits data once into training and testing sets, whereas k-fold cross-validation rotates the test set through k parts.
- **Kinyarwanda:**
Holdout validation igabanya amakuru rimwe mu matsinda y'imyitoto n'igerageza, mu gihe k-fold cross-validation ikagabanya mu bice k kandi buri gice kigakoreshwa nk'igerageza rimwe.

145. What is overfitting, and how can it be detected during model evaluation?

- **English:**
Overfitting is when a model performs significantly better on training data than on testing data; it is detected by a large gap between training and testing performance.
- **Kinyarwanda:**
Overfitting iba iyo model ikora cyane ku makuru y'imyitoto kurusha ku makuru y'igerageza; ibi bigaragara iyo habayeho itandukaniro rinini hagati ya training na testing.

146. Describe the role of confusion matrices in evaluating classification models.

- **English:**
Confusion matrices provide a detailed breakdown of correct and incorrect predictions, facilitating error analysis.
- **Kinyarwanda:**
Confusion matrix igaragaza uko model yitwaye mu by'ukuri, igafasha mu gusuzuma amakosa mu gutahura ibyiciro.

147. What is precision, and how is it calculated from a confusion matrix?

- **English:**
Precision is calculated as $\frac{TP}{(TP+FP)}$ representing the fraction of correctly predicted positive instances.
- **Kinyarwanda:**
Precision ipimwa nka $\frac{TP}{(TP+FP)}$ ikerekana igipimo cy'ukuri mu byavuzwe ko ari positives.

148. Define recall and explain its importance in imbalanced datasets.

- **English:**
Recall is $\frac{TP}{TP+FN}$ and is crucial in imbalanced datasets to ensure that most actual positive cases are detected.
- **Kinyarwanda:**
Recall ni $\frac{TP}{TP+FN}$ kandi ni ingenzi cyane mu makuru adahwanye neza kugira ngo hamenyekane positives zose zishoboka.

149. What is the F1-score, and how is it used to evaluate models?

- **English:**
The F1-score is the harmonic mean of precision and recall, balancing the two metrics; it is especially useful when classes are imbalanced.
- **Kinyarwanda:**
F1-score ni impuzandengo ya harmonic ya precision na recall, ikaba ikoreshwa cyane mu gihe ibyiciro bitaba bingana.

150. How does the ROC curve help in evaluating binary classification models?

- **English:**
The ROC curve plots the true positive rate against the false positive rate at various thresholds, helping visualize the trade-off between sensitivity and specificity.
- **Kinyarwanda:**
ROC curve yerekana uko true positive rate ihagaze ugereranyije na false positive rate mu bipimo bitandukanye, bigafasha kureba uko sensitivity na specificity bihurira.

151. Explain the concept of false positives and false negatives.

- **English:**
False positives are instances where the model incorrectly predicts the positive class, while **false negatives** are instances where it fails to detect the positive class.
- **Kinyarwanda:**
False positives ni aho model ivuga ikintu nk'icy'ukuri ariko kitabaho, naho **false negatives** ni aho model itabasha kubona ibyo bikwiye kuba positives.

152. What are true positives and true negatives in classification problems?

- **English:**
True positives are cases correctly identified as positive, and true negatives are cases correctly identified as negative.
- **Kinyarwanda:**
True positives ni aho model imenye neza positives, naho true negatives ni aho imenye neza negatives.

153. When would you prioritize using mean absolute error (MAE) over mean squared error (MSE)?

- **English:**
MAE is preferred when outliers should not overly influence the error metric, as it gives a linear penalty for errors.
- **Kinyarwanda:**
MAE ikunda gukoreshwa iyo outliers zidashaka kugira uruhare runini mu ipimwa ry'amakosa, kuko itanga igihano cy'imibare mu buryo bworoshye.

154. How do you interpret an R-squared value close to 1 in a regression model?

- **English:**
An R-squared value near 1 indicates that the model explains most of the variance in the dependent variable.
- **Kinyarwanda:**
R-squared hafi ya 1 isobanura ko model isobanura neza impinduka z'icyo igamije.

155. What is cross-entropy loss, and when is it used in model evaluation?

- **English:**
Cross-entropy loss is a measure used primarily for classification problems, quantifying the difference between predicted probabilities and actual labels.
- **Kinyarwanda:**
Cross-entropy loss ni igipimo gikunze gukoreshwa mu gutahura ibyiciro, kikapima itandukaniro riri hagati ya probability ivuzwe n'iby'ukuri.

156. How does the AUC-ROC score indicate model performance?

- **English:**
The AUC-ROC score measures the area under the ROC curve; a higher score indicates better discriminative ability of the model.
- **Kinyarwanda:**
AUC-ROC score ipima umwanya uri muni ya ROC curve; agaciro kanini kerekana ko model ishoboye gutandukanya ibyiciro neza.

157. Describe the role of residual plots in regression model evaluation.

- **English:**
Residual plots display the differences between actual and predicted values, helping diagnose issues like non-linearity or heteroscedasticity in the model.
- **Kinyarwanda:**
Residual plots zerekana itandukaniro hagati y'ibivugwa n'iby'ukuri, bigafasha mu gusuzuma niba model ifite ibibazo nka non-linearity cyangwa heteroscedasticity.

158. How do you detect model bias in evaluation metrics?

- **English:**
Model bias is detected by comparing performance metrics across different subgroups or data segments to see if certain groups are systematically mispredicted.
- **Kinyarwanda:**
Model bias iboneka iyo ugereranyije metrics mu matsinda atandukanye y'amakuru, ukabona ko hari itsinda runaka ritabasha gutahurwa neza.

159. What is the importance of validation data in model evaluation?

- **English:**
Validation data is used to fine-tune model parameters and prevent overfitting by providing an unbiased performance estimate during training.
- **Kinyarwanda:**
Amakuru ya validation akoreshwa mu guhindura parameters no gukumira overfitting, kuko atanga igipimo cy'ukuri ku buryo model ikora.

160. How can you improve model performance based on evaluation results?

- **English:**
Improvements can include **hyperparameter tuning**, **feature engineering**, **collecting more data**, and possibly selecting different algorithms.
- **Kinyarwanda:**
Ushobora kunoza model ukoresheje guhindura hyperparameters, guhitamo features neza, gukusanya amakuru menshi, cyangwa guhitamo algorithm itandukanye.

Unit 2.5: Tuning Hyperparameters

161. What is hyperparameter tuning in machine learning?

- **English:**
Hyperparameter tuning is the process of selecting the best set of hyperparameters (like learning rate, batch size) to optimize model performance.
- **Kinyarwanda:**
Hyperparameter tuning ni uburyo bwo guhitamo hyperparameters nziza (nko learning rate, batch size) kugira ngo model ikore neza.

162. Explain the difference between model parameters and hyperparameters.

- **English:**
Model parameters are learned during training (e.g., weights in a neural network), while **hyperparameters** are set before training and govern the learning process.
- **Kinyarwanda:**
Parameters za model zibonerwa mu myitozo (nko weights muri neural network), naho **hyperparameters** zibanziriza imyitozo zikagena uko model yiga.

163. What is grid search, and how is it used for hyperparameter tuning?

- **English:**
Grid search is a systematic method for exploring a predefined set of hyperparameters to find the combination that yields the best performance.
- **Kinyarwanda:**
Grid search ni uburyo bwo kugerageza uburyo butandukanye bwa hyperparameters bugamije kubona imiterere itanga ibisubizo byiza.

164. Describe the random search technique for hyperparameter optimization.

- **English:**
Random search randomly samples hyperparameter combinations rather than exhaustively testing every possibility, often yielding good results with less computation.
- **Kinyarwanda:**
Random search itoranya hyperparameter mu buryo butunguranye aho kugerageza buri buryo bushoboka, bityo igatanga ibisubizo byiza mu gihe gito.

165. What are the advantages of using Bayesian optimization in hyperparameter tuning?

- **English:**
Bayesian optimization uses a probabilistic model to predict promising hyperparameters, reducing the number of evaluations required to find optimal settings.
- **Kinyarwanda:**
Bayesian optimization ikoresha probability model mu guteganya hyperparameters nziza, bityo igabanya umubare w'ibigeragezo kugira ngo ubone configuration ikwiye.

166. How does cross-validation assist in hyperparameter tuning?

- **English:**
Cross-validation provides a robust estimate of model performance for each hyperparameter configuration, ensuring that the selected parameters generalize well.
- **Kinyarwanda:**
Cross-validation itanga igipimo cy'ukuri cya performance ya model kuri buri hyperparameter configuration, bityo ikemeza ko parameters zahiswemo zikora neza ku makuru yose.

167. What is the learning rate, and why is it a critical hyperparameter?

- **English:**
The learning rate controls the magnitude of parameter updates during gradient descent; if set too high, the model may diverge, and if too low, training becomes slow.
- **Kinyarwanda:**
Learning rate igenzura ingano y'intambwe mu guhindura parameters muri gradient descent; niba ari nyinshi cyane bishobora gutuma model itikanga, naho niba ari nke bituma imyitozo iba itinze.

168. Describe how batch size impacts model training during hyperparameter tuning.

- **English:**
Batch size determines how many training samples are processed before updating the model; it affects the stability and speed of convergence.
- **Kinyarwanda:**
Batch size igena umubare w'ibyitegererezo bikoreshwa mbere yo kuvugurura model; bigira uruhare mu gutuma convergence iba ikomeye kandi byihuta.

169. What is regularization, and how do hyperparameters like L1 and L2 affect it?

- **English:**
Regularization adds a penalty term to the loss function to prevent overfitting; L1 and L2 hyperparameters control the strength of this penalty.
- **Kinyarwanda:**
Regularization yongeza penalty kuri loss function kugira ngo hirindwe overfitting; L1 na L2 ni hyperparameters zigenzura uko penalty iba nini.

170. Explain the importance of tuning the number of hidden layers in neural networks.

- **English:**
The number of hidden layers affects the network's capacity to learn complex patterns; too few may lead to underfitting, while too many can cause overfitting.
- **Kinyarwanda:**
Umubare w'amayeri afunze (hidden layers) ugenzura ubushobozi bwa neural network bwo kumenya imiterere ikomeye; nke cyane bishobora gutera underfitting, naho nyinshi cyane bishobora gutera overfitting.

171. What is dropout, and how is it used as a regularization technique?

- **English:**
Dropout randomly deactivates a percentage of neurons during training, preventing the network from relying too much on any one neuron and reducing overfitting.
- **Kinyarwanda:**
Dropout ni uburyo bwo guhagarika ku buryo butunguranye igipimo runaka cya neurons mu myitozo, bigatuma network idatererana cyane kuri neuron imwe, bityo bigafasha kugabanya overfitting.

172. Describe the role of the activation function as a hyperparameter.

- **English:**
The activation function introduces non-linearity into the model; selecting an appropriate activation (e.g., ReLU, sigmoid) is crucial for model performance.
- **Kinyarwanda:**
Activation function ituma model iba non-linear; guhitamo activation ikwiriye (nko ReLU cyangwa sigmoid) ni ingenzi cyane ku musaruro

173. What are common hyperparameters in decision tree models?

- **English:**
Common hyperparameters include `max_depth`, `min_samples_split`, and `min_samples_leaf`.
- **Kinyarwanda:**
Hyperparameters zisanzwe muri decision trees ni `max_depth`, `min_samples_split`, na `min_samples_leaf`.

174. How does the `max_depth` hyperparameter affect a decision tree's performance?

- **English:**
`max_depth` limits the depth of the tree; a shallow tree may underfit, whereas a very deep tree may overfit.
- **Kinyarwanda:**
`max_depth` igena uko igiti cya decision tree kirekire; gito cyane gishobora gutera underfitting naho kirekire cyane kikaba overfitting.

175. Explain the role of `n_estimators` in ensemble models like Random Forest.

- **English:**
`n_estimators` specifies the number of trees in the ensemble; generally, more trees lead to improved performance but increase computational cost.
- **Kinyarwanda:**
`n_estimators` ni umubare w'ibiti bikoreshwa mu itsinda rya Random Forest; ibiti byinshi bikunze gutanga ibisubizo byiza ariko bisaba computation nyinshi.

176. What is early stopping, and how can it be used during hyperparameter tuning?

- **English:**
Early stopping halts training when performance on validation data ceases to improve, thereby preventing overfitting during hyperparameter tuning.
- **Kinyarwanda:**
Early stopping ni uburyo bwo guhagarika imyitozo igihe performance ku validation data itakomeza kunozwa, bigafasha kwirinda overfitting mu gihe cyo guhindura hyperparameters.

177. Describe the role of momentum in gradient descent optimization.

- **English:**
Momentum uses previous gradients to smooth out updates during gradient descent, leading to faster convergence and reducing oscillations.
- **Kinyarwanda:**
Momentum ikoresha gradients zabanje mu gufasha gukosora updates mu gradient descent, bigatuma convergence iba vuba kandi bigafasha kugabanya guhindagurika.

178. What is the impact of tuning the dropout rate in deep learning models?

- **English:**
Adjusting the dropout rate changes the fraction of neurons dropped during training; a higher rate can reduce overfitting but may also slow learning if too high.
- **Kinyarwanda:**
Guhindura dropout rate bigena igipimo cya neurons zihagarikwa mu myitoto; rate iri hejuru cyane ishobora kugabanya overfitting ariko ikanatuma imyitoto iba itinze niba yararenze urugero.

179. How does hyperparameter tuning differ between supervised and unsupervised models?

- **English:**
In supervised models, hyperparameter tuning focuses on improving prediction accuracy, whereas in unsupervised models it focuses on enhancing clustering quality or representation learning.
- **Kinyarwanda:**
Mu supervised models, hyperparameter tuning igamije kongera ukuri mu gutahura, mu gihe mu unsupervised models igamije kunoza clustering cyangwa representation learning.

180. Why is it important to avoid overfitting during hyperparameter tuning?

- **English:**
Overfitting during tuning results in a model that performs exceptionally on training data but poorly on new data, defeating the purpose of generalization.
- **Kinyarwanda:**
Kwiga cyane ku makuru y'imyitoto mu gihe cyo guhindura hyperparameters bituma model ikora neza ku makuru y'imyitoto ariko ikitwara nabi ku makuru mashya, bigatuma itabasha gusobanukirwa neza.

Unit 3.1: Description of Model Deployment Methods

181. What is model deployment in machine learning?

- **English:**
Model deployment is the process of integrating a trained model into a production environment so that it can make real-time predictions.
- **Kinyarwanda:**
Model deployment ni ugushyira model yatojwe mu buzima busanzwe (production) kugira ngo ishobore gutanga ibisubizo mu gihe nyacyo.

182. Why is model deployment an important step in the machine learning lifecycle?

- **English:**
It transforms a trained model into a practical application, allowing businesses or systems to use the predictions for decision making.

- **Kinyarwanda:**
Ibi bituma model yatojwe iba igikoreshwa mu buzima busanzwe, bityo ikaba ifasha mu gufata imyanzuro mu bucuruzi cyangwa mu mikorere y'ibindi bisabwa.

183. List three common methods for deploying machine learning models.

- **English:**
 1. Batch deployment
 2. Real-time (online) deployment
 3. Model-as-a-Service (MaaS)
- **Kinyarwanda:**
 1. Batch deployment
 2. Real-time deployment
 3. Model-as-a-Service (MaaS)

184. What is the difference between batch deployment and real-time deployment?

- **English:**
Batch deployment processes data in groups at scheduled intervals, while real-time deployment processes data immediately as it is received.
- **Kinyarwanda:**
Batch deployment ikora ku makuru mu matsinda ku gihe cyateganijwe, naho real-time deployment ikora ku makuru ako kanya igihe agera.

185. Explain the term "Model-as-a-Service" (MaaS).

- **English:**
MaaS refers to the delivery of machine learning models over the cloud as a service accessible via APIs.
- **Kinyarwanda:**
MaaS bivuze gutanga models za ML nka service yo mu bushorishori abasha kugerwaho binyuze kuri APIs.

186. How does containerization support model deployment?

- **English:**
Containerization packages the model with its dependencies (using tools like Docker), ensuring consistency and portability across environments.
- **Kinyarwanda:**
Containerization ipfunyika model hamwe n'ibyayo byose (ikoresheje nka Docker), bigatuma deployment iba imwe kandi ishobora gukoreshwa ahantu hatandukanye.

187. What is Docker, and how is it used in machine learning deployment?

- **English:**
Docker is a platform that **automates the deployment** of applications inside **containers**; in ML, it is used to **package models** and their environments for **consistent deployment**.
- **Kinyarwanda:**
Docker ni urubuga rwo gushyira mu bikorwa applications mu macontainers; mu ML, ikoresheya mu gupfunyika models n'ibidukikije byazo kugira ngo deployment ibe imwe.

188. Explain the concept of a RESTful API in the context of model deployment.

- **English:**
A **RESTful API** uses standard **HTTP** methods to allow external applications to communicate with and obtain predictions from a deployed model.
- **Kinyarwanda:**
RESTful API ikoresheya uburyo busanzwe bwa **HTTP** kugira ngo porogaramu zindi zishobore kuvugana na model yashyizwe mu bikorwa no kubona ibisubizo.

189. What are the advantages of deploying models using cloud services?

- **English:**
Cloud deployment offers **scalability, flexibility**, ease of maintenance, and seamless integration with other cloud-based tools and services.
- **Kinyarwanda:**
Deployment mu bushorishori butanga **scalability, flexibility**, korohera maintenance, ndetse no guhuza n'ibindi bikoresho byo mu bushorishori.

190. Name three popular cloud platforms used for machine learning deployment.

- **English:**
 1. AWS (Amazon Web Services)
 2. Google Cloud Platform
 3. Microsoft Azure
- **Kinyarwanda:**
 1. AWS
 2. Google Cloud Platform
 3. Microsoft Azure

191. What is the role of AWS SageMaker in model deployment?

- **English:**
AWS SageMaker is an end-to-end service that facilitates building, training, and deploying machine learning models in the cloud.
- **Kinyarwanda:**
AWS SageMaker ni serivisi yuzuye ituma bikorwa byose byo gukora, gutoza, no gushyira mu bikorwa models za ML mu bushorishori.

192. Describe the process of deploying a machine learning model with Flask.

- **English:**
With **Flask**, a lightweight web framework, you can create an API endpoint that loads the trained model and processes incoming requests to return predictions.
- **Kinyarwanda:**
Ukoresheje Flask, ushobora gukora API endpoint ishyiraho model yatojwe igakira amakuru maze ikagaragaza ibisubizo.

193. What is Kubernetes, and how does it assist in managing deployed models?

- **English:**
Kubernetes is an **orchestration platform for containers** that **automates deployment, scaling, and management**, ensuring that deployed models remain available and can be updated easily.
- **Kinyarwanda:**
Kubernetes ni uburyo bwo gucunga **containers** mu buryo bwikora, bugafasha mu gushyira mu bikorwa, kongera cyangwa kugabanya, no gucunga models za ML mu buryo bwizewe.

194. Explain the term CI/CD and its relevance to model deployment.

- **English:**
CI/CD (Continuous Integration/Continuous Deployment) automates testing and deployment pipelines, ensuring that new model versions are seamlessly integrated into production.
- **Kinyarwanda:**
CI/CD (Continuous Integration/Continuous Deployment) ituma testing na deployment bikorwa mu buryo bwikora, bigatuma version nshya za model zishyirwa mu bikorwa neza.

195. What are the key components of a CI/CD pipeline for machine learning?

- **English:**
Key components include source code management, **automated testing**, model training, and **automated deployment**.
- **Kinyarwanda:**
Ibice by'ingenzi ni gucunga code, automated testing, gutoza model, no gushyira mu bikorwa mu buryo bwikora.

196. How does model versioning improve deployment processes?

- **English:**
Model versioning tracks changes to the model over time, enabling rollbacks to previous versions if the new version underperforms, ensuring reliability in production.
- **Kinyarwanda:**
Model versioning ikurikirana impinduka za model uko ziteye, bigafasha gusubira kuri

version ishaje iyo version nshya itakora neza, bityo ikongera umutekano mu buzima busanzwe.

197. What is model drift, and why is it important to monitor deployed models for drift?

- **English:**
Model drift occurs when the model's predictive performance deteriorates over time due to changes in data patterns; monitoring drift is essential to trigger retraining or updates.
- **Kinyarwanda:**
Model drift iba iyo performance ya model igabanuka kubera impinduka mu miterere y'amakuru; gukurikirana drift ni ingenzi kugira ngo hamenyekane igihe cyo gutoza cyangwa kuvugurura.

198. Describe how model performance can degrade over time after deployment.

- **English:**
Over time, changes in data distribution, evolving trends, or outdated features can cause a model's performance to decline, necessitating retraining or updates.
- **Kinyarwanda:**
Mu gihe runaka, impinduka mu miterere y'amakuru, imigenzo mishya, cyangwa features zatakigira akamaro bishobora gutuma performance ya model igabanuka, bigasaba retraining cyangwa updates.

199. What is A/B testing in model deployment?

- **English:**
A/B testing involves comparing two model versions (A and B) in production to determine which one performs better under real-world conditions.
- **Kinyarwanda:**
A/B testing ni uburyo bwo kugereranya version ebyiri za model (A na B) mu bikorwa kugira ngo hamenyekane iyikora neza mu buzima busanzwe.

200. How does blue-green deployment reduce downtime during updates?

- **English:**
Blue-green deployment maintains two identical environments (blue and green); one environment serves production traffic while the other is updated, then traffic is switched, minimizing downtime.
- **Kinyarwanda:**
Blue-green deployment igira environments ebyiri zisa (blue na green); imwe ikora mu buryo busanzwe naho indi ivugururwa, hanyuma traffic ikoherewe ku version nshya, bigatuma downtime igabanuka.

Machine Learning – Questions 201 to 300 with Detailed Answers (English & Kinyarwanda)

Unit 3.2: Integration of Model File

201. What does it mean to integrate a machine learning model?

- **English:**
Integration means incorporating a trained machine learning model into an application or system so that it can be accessed and used in production.
- **Kinyarwanda:**
Guhuza model bisobanura gushyira model yatojwe mu ishyirahamwe cyangwa muri system kugira ngo ishobore gukoreshwa mu buzima busanzwe.

202. Why is model integration important in production environments?

- **English:**
It ensures that the insights and predictions from the model can be applied to real-world processes, supporting business decisions and automated systems.
- **Kinyarwanda:**
Ibi bifasha ko ibisubizo byavuye muri model bishobora gukoreshwa mu buzima busanzwe, bigafasha mu gufata imyanzuro y'ubucuruzi ndetse no mu gutunganya sisitemu zikorera mu buryo bwikora.

203. Explain the difference between model integration and model deployment.

- **English:**
Model deployment is the process of making the model available for use (e.g., via an API), whereas **model integration** focuses on connecting that model with other systems or applications.
- **Kinyarwanda:**
Model deployment ni ugushyira model mu bikorwa (nko kuyishyira kuri API), naho **model integration** ni uguhuza iyo model n'izindi sisitemu cyangwa porogaramu.

204. What are common file formats used to save machine learning models?

- **English:**
Common formats include **Pickle** (.pkl), **Joblib** (.joblib), **HDF5** (.h5), **ONNX** (.onnx), and **JSON** (for model configurations).
- **Kinyarwanda:**
Imiterere isanzwe ikoreshwa ni nka **Pickle** (.pkl), **Joblib** (.joblib), **HDF5** (.h5), **ONNX** (.onnx), ndetse na **JSON** (ku bisobanuro bya model).

205. Describe how to save a trained model using Python's pickle library.

- **English:**
You can use the pickle module as follows:

```
import pickle
with open('model.pkl', 'wb') as f:
    pickle.dump(trained_model, f)
```

- **Kinyarwanda:**
Ushobora gukoresha pickle utya:

```
import pickle
with open('model.pkl', 'wb') as f:
    pickle.dump(trained_model, f)
```

206. What is the role of ONNX (Open Neural Network Exchange) in model integration?

- **English:**
ONNX provides an open format to represent deep learning models, enabling **interoperability** between different **frameworks** and tools during integration.
- **Kinyarwanda:**
ONNX itanga uburyo bw'ubwisanzure bwo guhagararira models za deep learning, bityo igahuza frameworks zitandukanye mu gihe cyo guhuza model n'izindi sisitemu.

207. How does serialization support model integration?

- **English:**
Serialization converts a model into a format that can be easily saved, transferred, and later loaded for inference, ensuring consistency across platforms.
- **Kinyarwanda:**
Serialization ihindura model mu buryo bushobora kubikwa, koherezwa, no kongera gukoreshwa mu gihe cyo gutanga ibisubizo, bigatuma ihuza neza muri platforms zitandukanye.

208. Explain how JSON is used in model integration.

- **English:**
JSON is often used to store and exchange configuration settings or lightweight model metadata between systems during integration.
- **Kinyarwanda:**
JSON ikunze gukoreshwa mu kubika no guhererekanya ibisobanuro by'imyirondoro ya model cyangwa configuration hagati ya sisitemu mu gihe cyo guhuza model.

209. What is the purpose of API endpoints when integrating machine learning models?

- **English:**
API endpoints serve as interfaces through which external applications can send data to the model and receive predictions.
- **Kinyarwanda:**
API endpoints ni imiryango aho porogaramu zindi zishobora kohereza amakuru kuri model no kwakira ibisubizo.

210. Describe how to integrate a machine learning model with a web application.

- **English:**
Integration typically involves developing an **API** (using frameworks such as Flask or **FastAPI**) that loads the model and processes HTTP requests from the web app to return predictions.
- **Kinyarwanda:**
Guhuza model na web application bisaba gukora API (ukoresheje frameworks nka Flask cyangwa FastAPI) ishyiraho model maze ikakira requests za HTTP kuva muri web app igasubiza ibisubizo.

211. What is an inference engine, and how is it used in model integration?

- **English:**
An inference engine is a system component that executes a trained model to produce predictions based on input data; it is central to integrating the model into production workflows.
- **Kinyarwanda:**
Inference engine ni igice cya sisitemu gishyira mu bikorwa model yatojwe kugira ngo itange ibisubizo hashingiwe ku makuru yinjijwe; ni ingenzi mu guhuza model mu mikorere ya production.

212. Explain the role of databases in model integration.

- **English:**
Databases store input data, prediction results, and sometimes model metadata, ensuring that data flows seamlessly between the application and the model.
- **Kinyarwanda:**
Databases zibika amakuru yinjira, ibisubizo bya model, ndetse rimwe na rimwe ibisobanuro bya model, bigatuma amakuru agenda neza hagati ya application na model.

213. What are the steps to connect a deployed model to a front-end application?

- **English:**
Steps include:
 1. **Creating API endpoints to expose the model's functionality.**
 2. **Implementing authentication and security.**

3. Connecting the front-end to these endpoints (e.g., via AJAX or fetch calls).
- **Kinyarwanda:**
Intambwe zirimo:
 1. Gukora API endpoints zigaragaza imikorere ya model.
 2. Gushyiraho uburyo bwo kwemeza umutekano (authentication).
 3. Guhuza front-end n'izo endpoints (nko gukoresha AJAX cyangwa fetch calls).

214. How do you ensure data compatibility when integrating machine learning models?

- **English:**
Data compatibility is ensured by standardizing data formats, using common serialization methods, and validating input/output structures between systems.
- **Kinyarwanda:**
Kwemeza ko amakuru ahuye bikorwa mu guhindura imiterere y'amakuru, gukoresha uburyo bumwe bwa serialization, no kugenzura ko structure y'amakuru yinjira n'asohoka ihuye.

215. What is data serialization, and why is it important in model integration?

- **English:**
Data serialization converts data or models into a format that can be easily stored or transmitted, which is crucial for saving models and sharing them across platforms.
- **Kinyarwanda:**
Data serialization ni uguhindura amakuru cyangwa models mu buryo bushobora kubikwa cyangwa koherezwa byoroshye, kandi ni ingenzi mu kubika no guhererekanya models hagati ya platforms.

216. Describe the use of REST APIs for integrating ML models with external applications.

- **English:**
REST APIs enable external applications to send requests and receive predictions from the ML model over HTTP, facilitating communication between diverse systems.
- **Kinyarwanda:**
REST APIs zifasha porogaramu zindi kohereza requests no kwakira ibisubizo bya model binyuze kuri HTTP, bigahuza sisitemu zitandukanye.

217. How can you test if a model has been correctly integrated?

- **English:**
You can test integration by sending known inputs through the API endpoints and verifying that the outputs match expected results, as well as using unit and integration tests.
- **Kinyarwanda:**
Ushobora kugerageza integration wohera amakuru azwi kuri API endpoints ukareba niba ibisubizo bihuye n'ibytezwe, ndetse ugakoreshe unit na integration tests.

218. What are some common integration issues, and how can they be resolved?

- **English:**
Common issues include **data format mismatches**, **authentication errors**, and **network latency**.

They can be resolved by enforcing data standards, using robust security protocols, and optimizing network performance.

- **Kinyarwanda:**
Ibibazo bisanzwe ni ukutahuza kw'imiterere y'amakuru, amakosa y'umutekano (authentication), n'umuvuduko muke w'itumanaho.

Ibi bishobora gukemurwa mu gushyiraho amabwiriza y'amakuru, gukoresha security protocols zikomeye, no kunoza performance y'itumanaho.

219. Explain how logging and monitoring tools help in model integration.

- **English:**
Logging and **monitoring tools** track **API requests**, **errors**, and **performance metrics**, making it easier to **detect issues**, **debug problems**, and **ensure reliable** model operation.
- **Kinyarwanda:**
Logging na **monitoring** bitanga uburyo bwo gukurikirana requests za API, amakosa, n'ibipimo by'umusaruro, bityo bigafasha kumenya ibibazo hakiri kare no gukosora.

220. What is the role of middleware in machine learning model integration?

- **English:**
Middleware acts as a **bridge** between the **model** and other **applications**, handling data **transformation**, **security**, and **communication protocols** to **ensure smooth integration**.
- **Kinyarwanda:**
Middleware ni umuhuza hagati ya **model** n'izindi **porogaramu**, ikora ku guhindura amakuru, umutekano, ndetse na protocols z'itumanaho, bigatuma integration ikora neza.

Unit 3.3: Delivering Prediction to the Clients

221. What does it mean to deliver machine learning predictions to clients?

- **English:**
It means **providing end users** (or client applications) with the outputs of a machine learning model, usually via APIs or **user interfaces**.
- **Kinyarwanda:**
Bisobanura gutanga ibisubizo bya model ku bakoresha cyangwa porogaramu z'abakiriya, akenshi hifashishijwe APIs cyangwa imbonerahamwe.

222. Why is delivering accurate predictions critical in machine learning applications?

- **English:**
Accurate predictions ensure reliable **decision-making**, **improve user trust**, and enhance the overall effectiveness of the application.
- **Kinyarwanda:**
Ibisubizo bifite ukuri bituma imyanzuro ifatwa neza, bigatera abakoresha kwizera sisitemu, ndetse bigatuma application ikora neza.

223. Explain the process of creating an API to deliver model predictions.

- **English:**
The process involves developing a **RESTful API using frameworks** (e.g., **Flask or FastAPI**) that loads the model, receives input data via **HTTP requests**, processes the data through the model, and returns the prediction as a response.
- **Kinyarwanda:**
Uburyo bukubiyemo gukora RESTful API ukoresheje frameworks (nka Flask cyangwa FastAPI) ishyiraho model, ikakira amakuru binyuze kuri HTTP, ikayasesengura na model maze igatanga ibisubizo.

224. What is the role of HTTP requests in delivering model predictions?

- **English:**
HTTP requests allow client applications to send input **data to the API** and receive prediction outputs in return, facilitating communication between the client and the model.
- **Kinyarwanda:**
HTTP requests zituma porogaramu z'abakiriya zishobora kohereza amakuru kuri API maze zikabona ibisubizo, bityo zikagirana itumanaho na model.

225. Describe the difference between GET and POST requests when delivering predictions.

- **English:**
GET requests are used to **retrieve data** (without modifying it), while

POST requests **send data to the server** (often used when providing input for predictions).
- **Kinyarwanda:**
GET requests zikoreshya mu gusaba amakuru (udahindura), naho **POST** requests zohereza amakuru kuri server (akenshi zikoreshya mu gutanga input ku model).

226. How do front-end applications interact with machine learning models to display predictions?

- **English:**
Front-end applications call API endpoints to send data, receive prediction responses, and then display the results using user interfaces such as dashboards or mobile screens.
- **Kinyarwanda:**
Front-end applications zihamagara API endpoints zo kohereza amakuru no kwakira ibisubizo, hanyuma bigasohorwa mu buryo bugaragara nko kuri dashboards cyangwa kuri mobile.

227. What is an API response, and what information does it typically contain?

- **English:**
An API response is the data sent back to the client after processing the request.

It typically contains prediction results, status codes, and sometimes metadata.
- **Kinyarwanda:**
API response ni amakuru atangwa ku mukiriya nyuma yo gutunganya request, akenshi akubiyemo ibisubizo bya prediction, status code, ndetse n'ibisobanuro by'inyongera.

228. How can JSON be used to structure model prediction outputs?

- **English:**
JSON structures data in a lightweight, human-readable format, making it easy to package predictions (e.g., as key-value pairs) for client consumption.
- **Kinyarwanda:**
JSON ibika amakuru mu buryo bworoshye gusoma kandi bworoheje, bigafasha gupfunyika ibisubizo bya model (nko muri key-value pairs) ku buryo abakiriya babibona.

229. What are the performance considerations when delivering real-time predictions?

- **English:**
Considerations include minimizing latency, ensuring high throughput, managing server load, and providing scalability to handle many requests.
- **Kinyarwanda:**
Ibintu by'ingenzi ni ukugabanya latency, gutanga throughput ihagije, gucunga umubare munini w'ibisabwa (server load), no guha system ubushobozi bwo kwaguka

230. Explain how caching can improve the performance of prediction delivery.

- **English:**
Caching stores **frequently requested** prediction results temporarily, reducing computation time and speeding up response times for repeated queries.
- **Kinyarwanda:**
Caching ibika by'igihe gito ibisubizo bya prediction bikoreshwa kenshi, bigabanya igihe cyo kubitunganya kandi bigatuma response iba vuba.

231. What are latency issues, and how can they affect prediction delivery?

- **English:**
Latency refers to delays between sending a request and receiving a response; high latency can slow down user experience and hinder real-time decision-making.
- **Kinyarwanda:**
Latency ni igihe gihita hagati yo kohereza request no kwakira ibisubizo; latency iri hejuru ishobora gutuma response itinda, bigatuma experience y'umukoresha igenda nabi.

232. Describe how load testing helps in optimizing prediction delivery.

- **English:**
Load testing simulates high traffic to evaluate how well the prediction service performs under stress, identifying bottlenecks and ensuring system robustness.
- **Kinyarwanda:**
Load testing itera kugerageza sisitemu mu bihe by'umuvuduko mwinshi, igafasha kumenya aho sisitemu itihanganira umubare munini w'amabwiriza no gutanga ibisubizo, bityo ikanoza imikorere yayo.

233. What is a client-server architecture in the context of prediction delivery?

- **English:**
It is an architectural model where the client (e.g., web or mobile app) sends requests to a server that hosts the machine learning model and returns predictions.
- **Kinyarwanda:**
Ni uburyo aho umukiriya (nko kuri web cyangwa mobile app) wohereza requests kuri server ifite model maze ikagarura ibisubizo.

234. How can WebSockets be used for real-time prediction updates?

- **English:**
WebSockets establish a **persistent connection** **WebSockets between the client and server**, allowing the server to push real-time prediction updates to the client without repeated requests.
- **Kinyarwanda:**
ishyiraho connection ihoraho hagati **y'umukiriya na server**, bigatuma server ishobora kohereza ibisubizo bya prediction ako kanya ku mukiriya idakeneye requests kenshi.

235. What is API throttling, and why is it important in prediction delivery?

- **English:**
API throttling limits the number of requests a client can make in a given time period, protecting the server from overload and ensuring fair usage.
- **Kinyarwanda:**
API throttling igabanya umubare w'amabwiriza umukiriya ashobora kohereza mu gihe runaka, bigafasha kurinda server gucika intege no gukoresha sisitemu neza.

236. Explain how error handling works when delivering predictions via APIs.

- **English:**
Error handling involves detecting failures during request processing and returning meaningful error messages and status codes so that clients can understand and resolve issues.
- **Kinyarwanda:**
Error handling isobanura uburyo bwo kumenya amakosa mu gutunganya requests no koherereza ubutumwa bwumvikana hamwe na status codes, bityo abakiriya bakabimenya kandi bakabikemura.

237. What is the purpose of logging predictions in production environments?

- **English:**
Logging captures prediction requests, responses, and errors to help monitor performance, debug issues, and audit system usage over time.
- **Kinyarwanda:**
Logging ibika amakuru y'ibisabwa, ibisubizo, n'amakosa, bigafasha gukurikirana uko sisitemu ikora, gukosora amakosa, no kugenzura uko ikoresheya.

238. How can monitoring tools help in maintaining prediction delivery services?

- **English:**
Monitoring tools track system metrics (e.g., response times, error rates) in real time, allowing for proactive maintenance and quick response to issues.
- **Kinyarwanda:**
Monitoring tools zikurikirana ibipimo bya sisitemu (nko gutinda kwa response, amakosa) mu gihe nyacyo, bigafasha mu kubungabunga sisitemu no gukemura ibibazo byihuse.

239. What are the security risks when delivering machine learning predictions to clients?

- **English:**
Risks include **data breaches**, **unauthorized access to APIs**, **injection attacks**, and **exposure of sensitive information**.
- **Kinyarwanda:**
Ibibazo bishobora kuba harimo kwinjira mu makuru atemewe, kwibasirwa kwa API, ibitero bya injection, no kugaragaza amakuru y'ibanga.

240. Describe how encryption can secure data during prediction delivery.

- **English:**
Encryption converts data into a coded format during transmission (using protocols like **HTTPS**), protecting it from interception or unauthorized access.

- **Kinyarwanda:**
Encryption ihindura amakuru mu buryo bw'ibanga mu gihe cyo kuyohereza (ukoreshwa **HTTPS**), ikayakingira ku gutwarwa cyangwa kwinjirirwa n'abantu batabyemerewe.

241. What is role-based access control (RBAC), and how is it applied to secure API endpoints?

- **English:**
RBAC restricts system access based on user roles, ensuring that only authorized users can access or modify API endpoints.
- **Kinyarwanda:**
RBAC ni uburyo bwo kugena uburenganzira hashingiwe ku mwanya w'umukoresha, bityo hakemzwa ko ari bo gusa bemerewe kugera kuri API endpoints cyangwa kuzikosora.

242. How does rate limiting help prevent abuse of prediction APIs?

- **English:**
Rate limiting restricts the frequency of requests, protecting the system from overload and malicious use by preventing excessive access.
- **Kinyarwanda:**
Rate limiting igabanya umubare w'amabwiriza umukiriya ashobora kohereza mu gihe runaka, bigafasha kurinda sisitemu gucika intege no gukumira ikoreshwa ry'uburiganya.

243. What are the best practices for documenting prediction APIs for client integration?

- **English:**
Best practices include providing **clear endpoint descriptions, request/response examples, error codes, authentication details, and usage guidelines.**
- **Kinyarwanda:**
Best practices zirimo gusobanura endpoints neza, gutanga ingero za requests/response, gusobanura error codes, amakuru y'umutekano, n'amabwiriza y'uko API ikoreshwa.

244. Explain how API gateways can improve prediction delivery.

- **English:**
API gateways manage, secure, and route **API traffic**, providing load balancing, authentication, and centralized monitoring for improved performance and security.
- **Kinyarwanda:**
API gateways zigenzura, zikingira, kandi zigahagararira traffic za API, zitanga load balancing, authentication, ndetse na centralized monitoring, bigatuma performance n'umutekano byiyongera.

245. What is the importance of API versioning in maintaining backward compatibility?

- **English:**
API versioning ensures that changes to the API do not break existing client integrations by allowing multiple versions to coexist, providing a smooth transition.
- **Kinyarwanda:**
API versioning yemeza ko impinduka zikorwa kuri API zidahungabanya uburyo abakiriya basanzwe bayikoresha, binyuze mu gutuma versions nyinshi zibaho icyarimwe mu buryo bworoshye.

246. How can predictive analytics dashboards improve user interaction with ML models?

- **English:**
Dashboards provide visual representations of prediction data, **trends**, and **performance metrics**, allowing users to easily understand, interact with, and make decisions based on model outputs.
- **Kinyarwanda:**
Predictive analytics dashboards zitanga amashusho y'amakuru, imiterere, n'ibipimo by'umusaruro bya model, bigafasha abakoresha kumva, gukorana, no gufata ibyemezo hashingiwe ku bisubizo bya model.

247. What is a feedback loop, and how does it improve model predictions over time?

- **English:**
A feedback loop collects user or system responses to predictions and feeds this information back into the model training process, leading to continual improvement.
- **Kinyarwanda:**
Feedback loop ni uburyo bwo gukusanya ibisubizo bitangwa n'abakoresha cyangwa sisitemu, bigasubizwa mu myitozo ya model, bityo ikarushaho kunozwa uko iminsi ihita.

248. Describe the process of updating deployed models based on client feedback.

- **English:**
The process involves monitoring prediction performance, gathering client feedback, retraining or fine-tuning the model with updated data, and redeploying the improved model.
- **Kinyarwanda:**
Ibi bikorwa mu gukurikirana uko model ikora, gukusanya ibisubizo by'abakiriya, kongera gutoza cyangwa guhindura model hakoreshejwe amakuru mashya, maze igashyirwa mu bikorwa bushya.

249. How does edge computing support real-time prediction delivery in IoT devices?

- **English:**
Edge computing processes data locally on devices rather than sending it to a centralized server, reducing latency and enabling real-time predictions in IoT environments.

- **Kinyarwanda:**
Edge computing ikora ku bikoresho ubwabyo aho kohereza amakuru kuri server imwe, bigatuma latency igabanuka kandi predictions zigerwaho mu gihe nyacyo muri IoT.

250. Explain how mobile applications can consume machine learning predictions.

- **English:**
Mobile apps consume predictions by sending user data to an API endpoint, receiving the prediction result, and displaying it in a user-friendly format on the device.
- **Kinyarwanda:**
Mobile applications zikoresha ibisubizo bya model mu kohereza amakuru y'umukoresha kuri API, zikakira ibisubizo, maze bikagaragazwa mu buryo bworoshye kuri telefoni.

Advanced Data Pre-processing and Practical Scenarios

251. What are the key differences between data wrangling and data cleaning?

- **English:**
Data cleaning focuses on **correcting errors**, **filling missing values**, and **removing noise**, whereas

data wrangling involves **reshaping** and **transforming** raw data into a structured format suitable for analysis.
- **Kinyarwanda:**
Data cleaning ireba gukosora amakosa, kuzuza aho amakuru abuze, no gukuraho urusaku, naho

data wrangling ikubiyemo guhindura imiterere y'amakuru atarutunganywa neza ngo abe ateguye neza.

252. Explain how data normalization improves the performance of machine learning algorithms.

- **English:**
Normalization scales numerical data to a common range, ensuring that features with larger scales do not dominate the learning process, thus improving model convergence and accuracy.
- **Kinyarwanda:**
Data normalization ishyira imibare ku gipimo gihwanye, bigatuma features zifite agaciro kanini zidahiga izindi mu myitozo, bityo bigatuma model yigira neza kandi igatanga ibisubizo byiza.

253. Describe the process of handling imbalanced datasets during data pre-processing.

- **English:**
Handling imbalanced datasets may involve techniques such as **oversampling** the **minority class**, **undersampling the majority class**, or **using synthetic data** generation (e.g., SMOTE) to **balance the data distribution**.
- **Kinyarwanda:**
Gukemura ikibazo cy'amakuru adahwanye bikorwa nko kongera iby'itsinda rito (oversampling), kugabanya iby'itsinda rinini (undersampling), cyangwa gukoresha synthetic data nka SMOTE kugira ngo amakuru abe ahwanye.

254. What techniques can be used to detect multicollinearity in a dataset?

- **English:**
Techniques include calculating the **Variance Inflation Factor (VIF)** for each feature and **analyzing correlation matrices to identify highly correlated variables**.
- **Kinyarwanda:**
Ushobora gukoresha **VIF (Variance Inflation Factor)** kuri buri feature ndetse no gusesengura correlation matrix kugira ngo ubone variables zifitanye isano ikabije.

255. How does feature engineering contribute to better model performance?

- **English:**
Feature engineering creates **new features** or **transforms existing ones** to **better capture the underlying patterns**, leading to **improved accuracy** and **model efficiency**.
- **Kinyarwanda:**
Feature engineering itunganya cyangwa ikongera features nshya zigamije gufata neza imiterere y'amakuru, bityo ikarushaho gutanga ibisubizo byiza no kunoza model.

256. What are dummy variables, and why are they used in data pre-processing?

- **English:**
Dummy variables are binary (0 or 1) representations of categorical variables, used to convert **non-numerical** data into a numerical format for modeling.
- **Kinyarwanda:**
Dummy variables ni uburyo bwo guhindura ibyiciro (categorical) bikagaragara nk'imibare 0 na 1, bityo amakuru adakoresha imibare akabasha gukoreshwa na model.

257. Describe the difference between label encoding and one-hot encoding.

- **English:**
Label encoding assigns a unique integer to each category, while one-hot encoding creates binary columns for each category, avoiding ordinal relationships.
- **Kinyarwanda:**
Label encoding itanga integer yihariye kuri buri cyiciro, naho one-hot encoding ikora inkingi zigaragaza buri cyiciro mu buryo bwa binary, ikirinda gushyiraho urutonde rw'ibyiciro.

258. When would you choose to use Min-Max scaling over Z-score normalization?

- **English:**
Min-Max scaling is used when you need to **rescale features to a fixed range (e.g., 0–1)**, whereas **Z-score normalization** is preferred when preserving the distribution's shape is important.
- **Kinyarwanda:**
Min-Max scaling ikoreshwa iyo ukeneye gushyira features ku gipimo runaka (nko 0–1), naho Z-score normalization ikenerwa iyo ushaka ko distribution iguma uko imeze.

259. How do missing values impact the training of a machine learning model?

- **English:**
Missing values can lead to **inaccurate model training, biased results**, and may even cause errors during the training process if not handled properly.
- **Kinyarwanda:**
Amakuru abuze ashobora gutuma model idatoza neza, ikagaragaza ibisubizo bifite ivangura, ndetse bikaba byatera amakosa mu myitozo.

260. Explain the importance of outlier detection in data pre-processing.

- **English:**
Detecting outliers is essential to prevent them **from skewing the model's learning process**, which can **result in reduced accuracy and misleading predictions**.
- **Kinyarwanda:**
Gutahura outliers ni ingenzi kugira ngo ntibihungabanye imyitozo ya model, bityo bigatuma ibisubizo bidahinduka cyangwa bitangwa nabi.

261. What are the advantages of using Principal Component Analysis (PCA) in data pre-processing?

- **English:**
PCA reduces dimensionality by transforming features into principal components, which helps **remove noise, reduce overfitting, and improve computational efficiency**.
- **Kinyarwanda:**
PCA igabanya dimensions ihindura features mu bice by'ingenzi (principal components), bigafasha gukuraho urusaku, kwirinda overfitting, no kunoza uko computation ikorwa.

262. Describe how data leakage can occur during the pre-processing phase.

- **English:**
Data leakage happens when information from the test set is inadvertently used during pre-processing (e.g., scaling, imputation), leading to overly optimistic performance estimates.
- **Kinyarwanda:**
Data leakage iba iyo amakuru y'igerageza akoreshwa mu gihe cyo gutunganya data (nko

gukoresha scaling cyangwa imputation), bigatera ko model itangwa ibisubizo by'igishuko.

263. What are the best practices for handling time-series data during pre-processing?

- **English:**
Best practices include maintaining temporal order, handling seasonality, dealing with missing timestamps, and avoiding data leakage by not shuffling data.
- **Kinyarwanda:**
Best practices zirimo kubika umwanya w'amakuru (temporal order), gufata neza seasonality, gukemura ikibazo cy'aho timestamps zibura, no kwirinda data leakage mu kudahindura gahunda y'amakuru.

264. Explain the concept of feature selection and its significance in model development.

- **English:**
Feature selection involves choosing the most relevant features for the model, reducing noise, improving accuracy, and decreasing computational cost.
- **Kinyarwanda:**
Feature selection ni uguhitamo features z'ingenzi cyane, bigatuma urusaku rugabanuka, ukuri kwiyongera, kandi bigatuma computation igenda neza.

265. How does correlation analysis help in identifying redundant features?

- **English:**
Correlation analysis examines the relationship between features; **highly correlated** features can be redundant, so one may be removed to simplify the model without losing significant information.
- **Kinyarwanda:**
Correlation analysis isuzuma uko features zihurira; iyo features zikomeye cyane mu guhurira, zishobora kuba zidakenerye kongera, bityo ushobora gukuraho imwe ngo model ibe yoroshye udatakaje amakuru akenewe.

Additional Questions for Unit 2: Advanced Algorithm Concepts and Practical Applications

266. What is the difference between bagging and boosting algorithms? Provide examples.

- **English:**
Bagging (Bootstrap Aggregating) trains multiple independent models and aggregates their predictions (e.g., Random Forest). Boosting trains models sequentially, focusing on mistakes of previous models (e.g., AdaBoost, Gradient Boosting).
- **Kinyarwanda:**
Bagging (Bootstrap Aggregating) yitoza models zitandukanye mu buryo budafitanye isano maze igahuza ibisubizo (nko Random Forest). Boosting itera models zikurikirana,

igashyira imbere amakosa ya models zabanje (nko AdaBoost cyangwa Gradient Boosting).

267. Explain the role of gradient descent in optimizing machine learning models.

- **English:**
Gradient descent is an iterative optimization technique that updates model parameters by moving in the direction of the steepest decrease in the loss function, thus minimizing error.
- **Kinyarwanda:**
Gradient descent ni uburyo bwo guhindura parameters za model mu ntambwe ntoya bugamije kugabanya loss function, bityo bugabanya amakosa.

268. How does feature importance help improve model interpretability?

- **English:**
Feature importance assigns scores to input features based on their impact on predictions, allowing users to understand which features drive the model's decisions.
- **Kinyarwanda:**
Feature importance itanga amanota kuri buri feature hashingiwe ku ngaruka ifite ku bisubizo, bigafasha kumenya ibyo model yishingikirizaho mu gufata imyanzuro.

269. Describe the differences between L1 and L2 regularization.

- **English:**
L1 regularization (Lasso) adds the absolute value of coefficients as a penalty, promoting sparsity. **L2 regularization (Ridge)** adds the squared value of coefficients, discouraging large weights but not necessarily creating sparsity.
- **Kinyarwanda:**
L1 regularization (Lasso) yongeza agaciro ka absolute y'imibare nk'igihano, ikarushaho gutuma zimwe zibura (sparse). **L2 regularization (Ridge)** yongeza square y'imibare, igatuma habaho kugabanya imbaraga ariko ntibishobore gutuma zimwe zibura.

270. What are the key components of a neural network architecture?

- **English:**
Key components include an **input layer**, one or more **hidden layers** with activation functions, and an output layer.
- **Kinyarwanda:**
Ibice by'ingenzi ni input layer, hidden layers imwe cyangwa nyinshi zifite activation functions, hamwe na output layer.

271. How does dropout regularization prevent overfitting in deep learning models?

- **English:**
Dropout randomly disables a fraction of neurons during training, preventing the network from relying too heavily on any single neuron, thereby reducing overfitting.
- **Kinyarwanda:**
Dropout ihagarika ku buryo butunguranye igice cya neurons mu myitozo, bigatuma network itizirika cyane kuri neuron imwe, bityo ikagabanya overfitting.

272. Explain the vanishing gradient problem and how it can be mitigated.

- **English:**
The vanishing gradient problem occurs when gradients become very small in deep networks, slowing training. It can be mitigated by using activation functions like ReLU, residual connections, or normalization techniques.
- **Kinyarwanda:**
Vanishing gradient ni ikibazo aho gradients ziba nto cyane mu networks ziremereye, bigatuma imyitozo itinda. Ibi bishobora gukemurwa hifashishijwe activation functions nka ReLU, residual connections, cyangwa normalization.

273. What is the purpose of using convolutional layers in CNNs?

- **English:**
Convolutional layers automatically extract spatial features from input data (especially images) by applying filters, enabling effective feature detection.
- **Kinyarwanda:**
Convolutional layers zifasha mu gutahura features z'ibyerekemo mu makuru (by'umwihariko amashusho) ukoresheje filters, bityo zigafasha mu kumenya neza amakuru.

274. Describe the role of pooling layers in convolutional neural networks.

- **English:**
Pooling layers reduce the spatial dimensions of the feature maps, decreasing computational load and helping to prevent overfitting by summarizing features.
- **Kinyarwanda:**
Pooling layers zigabanya dimensions z'amashusho (feature maps), bigatuma computation igenda neza kandi bigafasha kwirinda overfitting mu gusubiramo features.

275. What are the key differences between RNNs and LSTMs in sequence modeling?

- **English:**
RNNs (Recurrent Neural Networks) process sequential data but may struggle with long-term dependencies. **LSTMs (Long Short-Term Memory)** are a type of RNN that use gating mechanisms to better capture long-term relationships.
- **Kinyarwanda:**
RNNs zikoresheya mu gucunga data z'ikurikiranyanyigisho ariko zikaba zifite ikibazo

cyo kwibagirwa igihe kirekire. LSTMs ni ubwoko bwa RNN bukoresha gating mechanisms kugira ngo bubashe gufata neza amakuru y'igihe kirekire.

276. Explain the concept of transfer learning and when it is most beneficial.

- **English:**
Transfer learning involves taking a pre-trained model on one task and fine-tuning it for another, saving training time and leveraging learned features, especially when data is limited.
- **Kinyarwanda:**
Transfer learning ni ugufata model yatojwe ku kintu kimwe maze ukayihindura gukoreshwa ku kindi, bityo ukagabanya igihe cyo gutoza kandi ukunguka features zizewe, cyane cyane iyo amakuru ari make.

277. What is data augmentation, and how does it improve model performance?

- **English:**
Data augmentation artificially increases the size of the training dataset by applying transformations (e.g., rotations, flips) to the data, which helps the model generalize better.
- **Kinyarwanda:**
Data augmentation ni uburyo bwo kongera dataset y'imyitoto mu buryo bw'ubukorano (nko guhindura amashusho, kuzizunguruka, kuzihindura), bigafasha model gusobanukirwa neza.

278. Describe the process of hyperparameter tuning using Random Search.

- **English:**
Random search selects random combinations of **hyperparameters** from a predefined range and **evaluates model performance**, often finding good settings with less computational cost than exhaustive grid search.
- **Kinyarwanda:**
Random Search itoranya hyperparameters mu buryo butunguranye mu bice byagenwe maze igapima uko model ikora, bigatanga configuration nziza mu gihe gito ugereranyije na grid search.

279. How does early stopping improve the training efficiency of deep learning models?

- **English:**
Early stopping halts training when validation performance ceases to improve, saving time and preventing overfitting.
- **Kinyarwanda:**
Early stopping ihagarika imyitoto igihe performance ya validation itakomeza kunozwa, bigatuma igihe kigabanywa kandi hakirindwa overfitting.

280. What are the key evaluation metrics for regression models? Explain each.

- **English:**
 - **Mean Squared Error (MSE):** Average of squared differences between predicted and actual values.
 - **Mean Absolute Error (MAE):** Average of absolute differences between predictions and true values.
 - **R-squared (R^2):** Proportion of variance in the dependent variable explained by the model.
- **Kinyarwanda:**
 - **Mean Squared Error (MSE):** Impuzandengo y'itandukaniro rya kabiri hagati y'ibyo model ivuga n'iby'ukuri.
 - **Mean Absolute Error (MAE):** Impuzandengo y'itandukaniro ry'imibare hagati y'ibivugwa na byo n'iby'ukuri.
 - **R-squared (R^2):** Igipimo cy'uko impinduka mu kintu cyifuzwa zisobanurwa na model.

281. Explain the difference between hard voting and soft voting in ensemble learning.

- **English:**

Hard voting takes the majority class predicted by each model, while soft voting averages the predicted probabilities for each class to determine the final outcome.
- **Kinyarwanda:**

Hard voting ifata icyiciro kinini kivugwa na buri model, naho soft voting ikagenda ifata average y'amahirwe (probabilities) ya buri cyiciro kugira ngo hamenyekane icyemezo cya nyuma.

282. How does the k-means algorithm work, and what are its limitations?

- **English:**

K-means partitions data into k clusters by minimizing the distance between points and cluster centroids.

Limitations include **sensitivity to initial centroids, difficulty handling clusters of varying shapes, and the need to predefine k.**

- **Kinyarwanda:**

K-means igabanya amakuru mu matsinda k hashingiwe ku kugabanya intera hagati y'ibyitegererezo na centroids y'amatsinda.

Imbogamizi ni ukwishingikiriza cyane kuri centroids za mbere, kuba bigoranye kubona amatsinda afite imiterere itandukanye, n'aho k igomba kuba yamenyekanye mbere.

283. Describe the process of building a machine learning pipeline using Scikit-learn.

- **English:**

A pipeline in Scikit-learn sequentially applies transformations (e.g., scaling, encoding) and then fits a model, ensuring reproducibility and simplifying the process.

- **Kinyarwanda:**
Pipeline muri Scikit-learn ikurikirana uburyo bwo gutunganya data (nko scaling, encoding) maze igafata model, bigatuma ibikorwa biba byoroshye kandi byongera reproducibility.

284. What is the role of learning rate schedulers in model optimization?

- **English:**
Learning rate schedulers adjust the learning rate during training, often reducing it as training progresses to fine-tune the model and improve convergence.
- **Kinyarwanda:**
Learning rate schedulers zihindura learning rate mu gihe cyo gutoza, akenshi zikayigabanya uko imyitozo igenda, bigafasha model kunoza imyitozo no kugera ku musaruro mwiza.

285. Explain the importance of data shuffling before model training.

- **English:**
Data shuffling randomizes the order of the training data, ensuring that the model does not learn spurious patterns due to ordering and improves generalization.
- **Kinyarwanda:**
Data shuffling ishyira mu buryo butunguranye urutonde rw'amakuru y'imyitozo, bigatuma model idahugira ku miterere itari nyayo bitewe n'urutonde rw'amakuru, bityo ikaba yiga neza.

Additional Questions for Unit 3: Advanced Deployment Techniques and Real-World Scenarios

286. What are the key steps involved in deploying a machine learning model to production?

- **English:**
Key steps include **preparing the environment, integrating the model, containerizing the application, setting up CI/CD pipelines, monitoring performance, and ensuring security.**
- **Kinyarwanda:**
Intambwe z'ingenzi ni gutegura ibidukikije, guhuza model, gushyira application muri container, gukora CI/CD pipelines, gukurikirana performance, no kwemeza umutekano.

287. How does model monitoring help maintain the performance of deployed models?

- **English:**
Monitoring tracks model performance over time (e.g., via metrics and logs), detecting degradation, drift, or failures so that corrective actions (such as retraining) can be taken promptly.

- **Kinyarwanda:**
Model monitoring ikurikirana uko model ikora mu gihe, igapima metrics na logs, ikamenya igihe performance igabanutse cyangwa drift yabaye, maze hakabaho ibikorwa byo gukosora nka retraining

288. Describe the process of creating RESTful APIs for serving machine learning models.

- **English:**
The process involves writing code using frameworks (e.g., Flask, FastAPI) to create endpoints that load the model, accept HTTP requests with input data, process the data through the model, and return predictions in a structured format.
- **Kinyarwanda:**
Uburyo bukubiyemo kwandika code ukoresheje frameworks (nka Flask, FastAPI) kugira ngo hakorwe endpoints zishyiraho model, zikakira HTTP requests zirimo amakuru, maze zisubiza ibisubizo mu buryo bugaragara.

289. What are the benefits of using Docker containers for model deployment?

- **English:**
Docker containers package the model and its **dependencies into a portable unit**, ensuring consistency **across different environments, scalability, and ease of deployment**.
- **Kinyarwanda:**
Docker containers zifata model n'ibyayo byose bikenewe mu gice kimwe, bigatuma deployment iba imwe, ikaba scalable kandi yoroshye kuyishyira mu bikorwa mu bice bitandukanye.

290. Explain the difference between serverless deployment and traditional deployment.

- **English:**
Serverless deployment abstracts away server management by running code in response to events, whereas traditional deployment requires provisioning and managing servers manually.
- **Kinyarwanda:**
Serverless deployment ntigikeneye gucunga servers kuko code ikorwa mu gihe habaye event runaka, naho traditional deployment isaba guha servers configuration no kuzikurikirana mu buryo bw'umwuga.

291. How does Kubernetes help with the scalability of deployed machine learning models?

- **English:**
Kubernetes automates **container orchestration**, allowing models to scale horizontally (adding more containers) as demand increases while managing load balancing and failover.
- **Kinyarwanda:**
Kubernetes icunga containers mu buryo bwikora, igatuma models zishobora kwaguka

mu buryo bwo kongera containers igihe habayeho umubare munini w'amabwiriza, ikanagena load balancing na failover.

292. What are microservices, and how do they support machine learning model deployment?

- **English:**
Microservices are small, independent services that work together; they allow different components of an ML system (such as data processing, prediction, and logging) to be developed, deployed, and scaled separately.
- **Kinyarwanda:**
Microservices ni serivisi nto zikorana, zituma ibice bitandukanye bya sisitemu ya ML (nko gutunganya data, gutanga prediction, na logging) bishobora gukorwa, gushyirwa mu bikorwa, no kwagurwa ku giti cyabyo.

293. Describe the security risks associated with model deployment and how to mitigate them.

- **English:**
Risks **include unauthorized access, data breaches, injection attacks, and model theft. Mitigation strategies involve encryption, secure authentication, regular security audits, and robust network configurations.**
- **Kinyarwanda:**
Ibibazo birimo kwinjirirwa mu buryo butemewe, kwibwa amakuru, injection attacks, no kwibwa model. Ibyo bishobora gukemurwa ukoresheje encryption, authentication ikomeye, audits za security buri gihe, n'imitunganyirize ya network ikomeye.

294. What is model rollback, and when would you need to perform it?

- **English:**
Model rollback is reverting to a previous version of a model if the new version underperforms or causes issues, ensuring minimal disruption to services.
- **Kinyarwanda:**
Model rollback ni ugusubira kuri version ya model yashyizweho mbere niba version nshya itakora neza cyangwa itera ibibazo, bityo bigatuma services zikomeza gukora neza.

295. Explain the concept of canary deployment in the context of machine learning models.

- **English:**
Canary deployment involves rolling out a new model version to a small subset of users first to monitor its performance before a full-scale release, reducing risk.
- **Kinyarwanda:**
Canary deployment ni uburyo bwo gushyira version nshya ya model ku itsinda rito ry'abakoresha mbere yo kuyishyira ku rwego rusange, bityo bigatuma habaho kugenzura neza performance no kugabanya ingaruka.

296. How do you implement continuous integration (CI) for machine learning projects?

- **English:**
CI for ML involves **automating the testing and integration of code changes**, including data validation, model training, and **deployment scripts**, using **tools such as Jenkins, GitLab CI, or CircleCI**.
- **Kinyarwanda:**
CI muri ML isaba gushyiraho uburyo bwo kugerageza no guhuza impinduka za code, harimo validation y'amakuru, gutoza model, na deployment scripts, ukoresheje ibikoresho nka Jenkins, **GitLab CI, cyangwa CircleCI**.

297. What are model serving frameworks, and how do they simplify deployment?

- **English:**
Model serving frameworks (e.g., **TensorFlow Serving, TorchServe**) provide ready-to-use infrastructure to host and manage ML models, automating aspects such as scaling, versioning, and inference.
- **Kinyarwanda:**
Model serving frameworks (nka TensorFlow Serving, TorchServe) zitanga uburyo bwihuse bwo kwakira no gucunga models za ML, zigakuraho ibibazo byo kwagura, versioning, na inference.

298. Describe the role of edge computing in machine learning model deployment.

- **English:**
Edge computing processes data and runs models on local devices (or near the data source), reducing latency and bandwidth usage, which is especially beneficial for IoT applications.
- **Kinyarwanda:**
Edge computing ikora ku bikoresho hafi y'aho amakuru aboneka, igatuma itinda rya network rigabanuka kandi latency igahinduka nto, bityo bigaba byiza cyane ku mishinga ya IoT.

299. What is API throttling, and how does it protect deployed models from abuse?

- **English:**
API throttling limits the number of requests per time period from a client, preventing server overload, reducing the risk of denial-of-service attacks, and ensuring fair usage.
- **Kinyarwanda:**
API throttling igabanya umubare w'amabwiriza umukiriya ashobora kohereza mu gihe runaka, bigatuma server itahungabana, bigafasha kwirinda ibitero bya denial-of-service, kandi bigatanga uburyo bwo gukoresha sisitemu mu buryo bwiza.

300. Explain the importance of model explainability when delivering predictions to clients.

- **English:**
Model explainability provides transparency into how predictions are made, which is critical for building trust with clients, ensuring regulatory compliance, and facilitating troubleshooting and improvement.
- **Kinyarwanda:**
Model explainability itanga uburyo bwo gusobanukirwa uko ibisubizo bitangwa, bityo bigafasha mu gutera abakiriya icyizere, gukurikiza amategeko, no gufasha mu gusuzuma no kunoza model.

Machine Learning Questions 301 to 400 with Detailed Answers (English & Kinyarwanda)

Advanced Scenario-Based Questions (Cross-Unit)

301. You're tasked with building a predictive model for a hospital to predict patient readmissions. What type of machine learning algorithm would you choose and why?

- **English:**
I would choose a supervised learning classification algorithm such as logistic regression or a decision tree. These algorithms work well when the target (readmission: yes/no) is labeled, are relatively interpretable, and allow medical professionals to understand the key factors influencing readmissions.
- **Kinyarwanda:**
Nahitamo algorithm yo mu bwoko bwa supervised learning (classification) nka logistic regression cyangwa decision tree. Izi algorithm zikora neza iyo ikintu kigomba guteganywa (nko gusubiramo kw'abarwayi: yego/oya) gifite labels, zikanoroshya gusobanura, bigafasha abakozi b'ubuvuzi kumenya impamvu nyamukuru zituma habaho gusubira kwa barwayi.

302. A company wants to detect fraudulent transactions in real time. Which machine learning technique would be most suitable? Explain your reasoning.

- **English:**
Real-time fraud detection is best handled by supervised learning algorithms especially ensemble methods (like Random Forest or Gradient Boosting) or even deep learning models that can process streaming data.

These models can quickly learn patterns from historical transaction data and update predictions as new data arrives.
- **Kinyarwanda:**
Gutahura uburiganya mu gihe nyacyo bikorerwa neza n'algorithms za supervised

learning, cyane cyane ensemble methods nka Random Forest cyangwa Gradient Boosting, ndetse n'ama deep learning models ashobora gucunga amakuru yinjira mu buryo bw'ako kanya. Izi model zifasha kwiga ku miterere y'amakuru y'igihe cyashize maze zikajya zivugurura ibisubizo uko amakuru mashya agera.

303. While working on a machine learning project, you notice that your model performs well on training data but poorly on testing data. What could be the cause, and how would you address it?

- **English:**

This is a classic sign of overfitting, where the model has learned the training data—including noise too well.

To address it, I would consider techniques such as cross-validation, regularization (L1/L2), reducing model complexity, or gathering more diverse data.

- **Kinyarwanda:**

Iki ni ikimenyetso cya overfitting, aho model yiga amakuru y'imyitozo cyane (harimo n'urusaku).

Kugirango dukemure iki kibazo, nakoresha cross-validation, regularization (L1/L2), kugabanya imiterere ya model, cyangwa gushaka amakuru menshi atandukanye.

304. Imagine you've deployed a recommendation system, and users are reporting irrelevant suggestions. What steps would you take to diagnose and improve the system?

- **English:**

I would start by analyzing user feedback and logging data to identify patterns of errors. Then, I'd review the data preprocessing and feature engineering steps, check for data drift, revalidate the model's parameters, and possibly retrain the model with updated or more diverse data.

- **Kinyarwanda:**

Nahera ku gusesengura ibisubizo by'abakoresha n'amalog, kugira ngo menye aho model ikora nabi. Hanyuma, nasubira mu gutunganya data (preprocessing), guhitamo features, kugenzura niba hari impinduka mu makuru (data drift), gusuzuma parameters za model, ndetse no kongera gutoza model ukoresheje amakuru mashya cyangwa atandukanye

305. You're working on a self-driving car model, and the dataset contains missing GPS data for some routes. How would you handle these missing values?

- **English:**

I would use data imputation techniques—such as interpolating missing GPS coordinates based on nearby values or using a statistical method (mean or median of nearby points). In some cases, I might also discard routes with too many missing values if they cannot be reliably recovered.

- **Kinyarwanda:**
Nakoresha uburyo bwa data imputation nko guhuza amakuru abuze ya GPS hifashishijwe amakuru ari hafi aho cyangwa gukoresha impuzandengo (mean/median) y'ibyegegeranyo bihuriye. Mu gihe hari inzira ifite amakuru menshi abuze kandi adashobora gusubirwamo neza, nshobora kuyisiba.

306. A marketing team wants to predict customer churn. Which evaluation metric would be more critical: precision or recall? Justify your answer.

- **English:**
In churn prediction, recall is typically more critical because it is important to identify as many actual churners as possible, even if that means having some false positives. Missing potential churners could be more costly.
- **Kinyarwanda:**
Mu guteganya customer churn, recall ni ingenzi kuko ari ngombwa kumenya abarwayi bose bashobora kugenda, nubwo byavamo false positives. Kureka kumenya abo bagomba kugenda bishobora kugira ingaruka zikomeye.

307. You've been given a large unstructured dataset (text and images) for sentiment analysis. Describe the data preprocessing steps you'd take.

- **English:**
For text: perform tokenization, remove stop words, and apply stemming/lemmatization. For images: resize, normalize pixel values, and perform data augmentation if needed. Finally, structure the data into a format suitable for the chosen model.
- **Kinyarwanda:**
Ku byerekeye inyandiko: nakora tokenization, nkavanaho amagambo adafite umumaro (stop words), ndetse n'ugukora stemming/lemmatization. Ku mashusho: nakora resizing, normalization y'amabara, ndetse nkongera data (data augmentation) niba bikenewe. Hanyuma, amakuru akoranywe mu buryo bwizewe bwa model.

308. A financial company's fraud detection model needs to process data in real time. Would you deploy the model on the cloud or edge devices? Explain your choice.

- **English:**
I would deploy the model on edge devices if low latency and immediate response are critical. However, if centralized data processing and scalability are more important, cloud deployment would be preferred. Often, a hybrid approach is used.
- **Kinyarwanda:**
Nayishyira ku bikoresho bya edge niba gutanga ibisubizo mu gihe nyacyo ari ingenzi. Ariko, niba kugenzura amakuru mu buryo buhuriywe hamwe no kwaguka ari ingenzi, deployment mu bushorishori (cloud) yaba ihwanye. Kenshi usanga hakoreshewa uburyo bwombi (hybrid).

309. Your deployed model's performance has dropped significantly over the past three months. What could be the reasons, and how would you address them?

- **English:**
Possible reasons include data drift (changes in the input data distribution), model degradation, or external factors affecting the underlying patterns. I would retrain the model with recent data, monitor for drift continuously, and update features as necessary.
- **Kinyarwanda:**
Impamvu zishoboka zirimo data drift (impinduka mu miterere y'amakuru yinjira), kugabanuka k'umusaruro wa model, cyangwa impamvu zituruka hanze zishobora guhindura imiterere. Nakongera gutoza model n'amakuru mashya, nkagenzura drift buri gihe, kandi nkavugurura features niba bikenewe.

310. You're asked to improve an existing machine learning model's accuracy. Would you focus on hyperparameter tuning, feature engineering, or data augmentation? Why?

- **English:**
The choice depends on the current model's issues. Often, I would start with feature engineering to ensure the model is receiving the most relevant information. If the features are sound, then hyperparameter tuning and data augmentation can further improve accuracy.
- **Kinyarwanda:**
Icyo wahitamo giterwa n'aho model ihagaze. Kenshi naherera ku guhitamo features neza (feature engineering) kugira ngo model ibone amakuru akenewe. Niba features zihagije, nyuma nakora hyperparameter tuning cyangwa data augmentation kugira ngo ukuri kwiyongere.

Critical Thinking and Higher-Order Questions

311. How can ethical considerations affect the deployment of machine learning models in sensitive areas like healthcare or criminal justice?

- **English:**
Ethical considerations include ensuring fairness, avoiding biases, maintaining patient privacy or due process, and preventing discriminatory outcomes.

Neglecting ethics can lead to mistrust and legal consequences.

- **Kinyarwanda:**
Ibyerekeye imyitwarire myiza birimo kwemeza ko ibisubizo bidafite ivangura, kwirinda guhera ku makuru ataboneye, kubungabunga uburenganzira bwa buri wese (nko mu buvuzi cyangwa mu butabera), no kwirinda ingaruka z'ubusumbane. Kutitondera ibi bishobora gutera kutizerwa no kugira ingaruka mu mategeko.

312. What are the potential biases that can occur during data collection, and how do they impact machine learning models?

- **English:**
Biases may include sampling bias, selection bias, or measurement bias. These biases can

lead models to learn patterns that do not represent the real world, resulting in skewed predictions and unfair outcomes.

- **Kinyarwanda:**
Ivangua rishobora kuba nka sampling bias, selection bias, cyangwa measurement bias. Ibi bishobora gutuma model yiga imiterere idahwitse, ikatanga ibisubizo bituzuye cyangwa bitandukanye n'ukuri, bityo bigatera kutabaho kw'ubutabera.

313. Explain how overfitting in machine learning is similar to memorization, and why this can be a problem for model generalization.

- **English:**
Overfitting is like memorization because the model learns the training data—including noise—instead of understanding the underlying patterns. This results in poor performance on new data because the model cannot generalize well.
- **Kinyarwanda:**
Overfitting isa no kwibuka ibintu byose kuko model yiga amakuru y'imyitoto (harimo n'urusaku) aho gusobanukirwa imiterere nyamukuru. Ibi bituma model idashobora gusobanukirwa neza amakuru mashya.

314. Compare and contrast the effectiveness of ensemble learning methods like bagging and boosting in handling noisy datasets.

- **English:**
Bagging reduces variance by averaging predictions from multiple models, which helps smooth out noise. Boosting focuses on correcting errors by sequentially training models, which can sometimes overemphasize noise if not properly regularized.
- **Kinyarwanda:**
Bagging igabanya variance ikoresheje guhuza ibisubizo bya models nyinshi, bigafasha gukuraho urusaku. Boosting yo ihangana n'amakosa mu buryo bwikurikiranya, ariko ishobora gushyira imbere urusaku cyane niba itagenzuwe neza.

315. If a model achieves 99% accuracy on an imbalanced dataset, why might this not be a reliable measure of performance? What alternative metrics would you consider?

- **English:**
In imbalanced datasets, high accuracy can be misleading because the model may simply be predicting the majority class. Alternative metrics such as precision, recall, F1-score, and AUC-ROC provide a better evaluation of performance on minority classes.
- **Kinyarwanda:**
Ku makuru adahwanye, 99% accuracy ishobora kuba ihimbano kuko model ishobora gusa gutangaza icyiciro kinini. Ibipimo byiza ni precision, recall, F1-score, na AUC-ROC kugira ngo usuzume neza uko model ikora ku byiciro bito

316. Describe the ethical implications of using machine learning models in surveillance technologies. How can developers mitigate these risks?

- **English:**
Surveillance ML models raise privacy concerns, risk of mass surveillance, and potential misuse by authorities.

Developers can mitigate these risks by implementing transparency, ensuring data anonymization, and adhering to strict ethical guidelines and legal regulations.

- **Kinyarwanda:**
Gukoresha ML mu gucunga umutekano bishobora guteza impungenge z'ubuzima bwite, gufata amakuru y'abantu menshi, no gukoresha nabi n'inzege. Abateza imbere bashobora kubikumira bagira transparency, bagakora anonymization y'amakuru, kandi bakubahiriza amabwiriza y'imyitwarire n'amategeko.

317. How does explainable AI (XAI) help in building trust in machine learning systems, especially in critical decision-making processes?

- **English:**
XAI provides insights into how a model makes decisions, allowing users to understand and validate the model's reasoning. This transparency is crucial in high-stakes applications such as healthcare and finance.
- **Kinyarwanda:**
XAI itanga ibisobanuro by'uko model ifata imyanzuro, bigafasha abantu kumva no kugenzura impamvu model yagiye ifata icyemezo runaka. Ibi ni ingenzi cyane mu bikorwa bifite ingaruka zikomeye nka healthcare na finance.

318. In what ways can data privacy regulations (like GDPR) influence machine learning model development and deployment?

- **English:**
Regulations may restrict the type of data that can be collected or used, require anonymization, and mandate data protection measures. This influences the design, training, and deployment of ML models to ensure compliance and protect individual privacy.
- **Kinyarwanda:**
Amategeko nka GDPR ashobora kugena ubwoko bw'amakuru ashobora gukusanywa cyangwa gukoresha, asaba ko hakorwa anonymization, ndetse asaba ingamba zo kurinda amakuru. Ibi bigira ingaruka ku buryo models zitegurwa, zitozwa, ndetse zishyirwa mu bikorwa.

319. What are the trade-offs between model complexity and interpretability in machine learning? Provide examples.

- **English:**
Complex models (like deep neural networks) often **achieve higher accuracy** but are **harder to interpret**.

Simpler models (like linear regression) are more **interpretable** but may **not capture complex patterns**. The trade-off depends on the application's need for transparency versus performance.

- **Kinyarwanda:**
Models zikomeye (nko deep neural networks) zikunze gutanga ibisubizo byiza ariko bigorana gusobanura uko zafashe imyanzuro.
- **Models zoroshye** (nko linear regression) ziroroshye gusobanura ariko zishobora kudashobora gufata imiterere ikomeye. icyo uhitamo giterwa n'uko application isaba ko ibintu bisobanurwa neza cyangwa performance iba ingenzi.

320. Imagine you've built a perfect predictive model. What could still go wrong when deploying it in the real world?

- **English:**
Even a perfect model can fail due to issues such as **data drift, integration problems, real-time latency challenges, security vulnerabilities, or changes in the external environment that make the model outdated.**
- **Kinyarwanda:**
N'ubwo model yaba itunganye neza, ishobora guhura n'ibibazo nko guhinduka k'umiterere y'amakuru (data drift), ibibazo mu guhuza na systems, gutinda mu gihe nyacyo (latency), ibibazo by'umutekano, cyangwa impinduka mu buryo bw'aho ikora bituma model iba ishaje.

Advanced Technical Questions Deep Learning and Model Optimization

321. What is the role of the softmax function in multi-class classification problems?

- **English:**
The **softmax function** converts raw scores (logits) into probabilities that sum to 1, making it ideal for multi-class classification by indicating the likelihood of each class.
- **Kinyarwanda:**
Softmax ihindura scores (logits) mu buryo bwa probabilities zose zikarenga kuri 1, bityo ikaba ikwiriye mu byiciro byinshi kuko igaragaza amahirwe y'ikicro cyose.

322. Explain the architecture of a Convolutional Neural Network (CNN) and how it processes image data.

- **English:**
A CNN consists of convolutional layers that apply filters to extract features, pooling layers that reduce spatial dimensions, and fully connected layers for classification.
- It processes image data by learning hierarchical patterns from edges to complex structures.

- **Kinyarwanda:**
CNN igizwe na convolutional layers zikoresheya mu gutahura features ukoresheje filters, pooling layers zigabanya dimensions z'amashusho, na fully connected layers zikora classification.
- Ikoresheya amashusho mu kwiga imiterere kuva ku mirongo (edges) ijya ku bice bigoye.

323. Describe how Long Short-Term Memory (LSTM) networks improve upon traditional RNNs.

- **English:**
LSTMs use gating mechanisms (input, forget, and output gates) to preserve long-term dependencies and avoid vanishing gradients, making them more effective for sequential data.
- **Kinyarwanda:**
LSTMs zikoresheya uburyo bwa gating (input, forget, output gates) kugira ngo zibike amakuru y'igihe kirekire kandi zirinde ikibazo cya vanishing gradients, bityo zikaba zikora neza ku makuru akurikiranye (sequential).

324. What is gradient clipping, and why is it used in training deep learning models?

- **English:**
Gradient clipping limits the size of gradient updates during training to prevent exploding gradients, which can destabilize the learning process.
- **Kinyarwanda:**
Gradient clipping igena urugero ntarwo rwa gradients mu gihe cyo gutoza, bigatuma hirindwa "exploding gradients" zishobora kwangiza imyitozo ya model.

325. Explain the difference between L1 and L2 regularization and their effects on model coefficients.

- **English:**
L1 regularization (Lasso) adds the absolute value of coefficients as a penalty, promoting sparsity.
- L2 regularization (Ridge) adds the squared value of coefficients, encouraging smaller, distributed weights without necessarily creating sparsity.
- **Kinyarwanda:**
L1 regularization (Lasso) yongeza absolute value y'imibare nk'igihano, ikarushaho gutuma zimwe zibura (sparse).
- L2 regularization (Ridge) yongeza square y'imibare, igatuma habaho kugabanya imbaraga ariko ntibishobore gutuma zimwe zibura.

326. How does the Adam optimizer differ from stochastic gradient descent (SGD)?

- **English:**
Adam combines the advantages of two other extensions of SGD by adapting the learning

rate for each parameter and using estimates of first and second moments of the gradients, leading to faster convergence.

- **Kinyarwanda:**
Adam itandukana na SGD isanzwe kuko ihindura learning rate kuri buri parameter ndetse ikoresha estimates z'ibanze (first and second moments) za gradients, bigatuma convergence iba vuba.

327. What are residual networks (ResNets), and how do they address the vanishing gradient problem?

- **English:**
ResNets use shortcut (skip) connections that allow gradients to flow directly through the network, mitigating the vanishing gradient problem and enabling the training of much deeper networks.
- **Kinyarwanda:**
ResNets zikoresha shortcut connections (skip connections) zituma gradients zigera ku ntera nini muri network, bigafasha gukemura ikibazo cya vanishing gradient ndetse bigatuma bishoboka gutoza networks ndende cyane.

328. Describe the attention mechanism in neural networks and its role in natural language processing (NLP).

- **English:**
The attention mechanism allows the network to focus on the most relevant parts of the input sequence, improving performance in tasks such as translation and text summarization by weighting important words more heavily.
- **Kinyarwanda:**
Attention mechanism ituma network ibasha kwita cyane ku bice by'ingenzi by'inyandiko, bigafasha mu mirimo nka translation cyangwa text summarization aho amagambo akomeye aha agaciro kanini.

329. What is transfer learning, and how can it reduce the need for large datasets?

- **English:**
Transfer learning uses a model pre-trained on one task and fine-tunes it for another related task, thereby reducing the amount of data and training time required for the new task.
- **Kinyarwanda:**
Transfer learning ni ugufata model yatojwe kuri task imwe maze ukayikoresha ku yindi ifitanye isano, bityo ugakenera amakuru make n'igihe gito cyo gutoza kuri iyo task nshya.

330. Explain how autoencoders work and their applications in anomaly detection.

- **English:**
Autoencoders are neural networks that learn to compress data into a lower-dimensional representation and then reconstruct it.
- They can detect anomalies by identifying inputs that have a high reconstruction error.
- **Kinyarwanda:**
Autoencoders ni neural networks ziga gufata amakuru no kuyagabanya (compress) maze zikayongera kuyubaka (reconstruct).
- Zishobora gukoreshwa mu gutahura anomalies aho zisanga inputs zitubahiriza reconstruction error iri hejuru.

Advanced Model Deployment and MLOps Questions

331. What is the difference between model serving and model deployment in an ML pipeline?

- **English:**
Model deployment is the process of putting a trained model into production, while **model serving** refers to providing an interface (such as an API) that allows users or applications to request predictions from the deployed model.
- **Kinyarwanda:**
Model deployment ni ugushyira model yatojwe mu bikorwa, naho model serving ni ugutanga interface (nka API) ituma abakoresha cyangwa porogaramu babasha gusaba ibisubizo bya model.

332. Explain how continuous integration (CI) and continuous deployment (CD) pipelines benefit ML workflows.

- **English:**
CI/CD pipelines automate testing, integration, and deployment of code changes, ensuring that model updates are reliable, reproducible, and quickly delivered to production.
- **Kinyarwanda:**
CI/CD pipelines zituma testing, integration, no gushyira mu bikorwa impinduka za code bikorwa mu buryo bwikora, bigatuma updates za model ziba zizewe, zibasha gusubirwamo, kandi zigashyirwa mu bikorwa vuba.

333. Describe the process of deploying a machine learning model as a REST API using Flask.

- **English:**
The process involves writing a Flask application that loads the trained model, defines API endpoints to accept input data via HTTP requests, processes the data through the model, and returns predictions as JSON responses.
- **Kinyarwanda:**
Uburyo bukubiyemo kwandika application ya Flask ishyiraho model yatojwe, ikagena

API endpoints zikakira amakuru binyuze muri HTTP, ikayasesengura na model maze igatanga ibisubizo muri format ya JSON.

334. What are the advantages of using Kubernetes for deploying machine learning models at scale?

- **English:**
Kubernetes automates **container orchestration**, making it easier to **manage, scale, and update multiple model deployments** while providing load balancing and fault tolerance.
- **Kinyarwanda:**
Kubernetes icunga containers mu buryo bwikora, bigatuma byoroha gucunga, kwagura, no kuvugurura models nyinshi, kandi itanga load balancing ndetse na fault tolerance.

335. Explain how Docker containers help achieve reproducibility in machine learning deployment.

- **English:**
Docker containers package the model along with its environment and dependencies, ensuring that the model runs identically across different systems and eliminating “it works on my machine” issues.
- **Kinyarwanda:**
Docker containers zifata model n’ibidukikije byayo byose, bigatuma model ikora kimwe ku mashini atandukanye, bityo bigakuraho ikibazo cyo kuvuga ngo “ikora kuri machine yanjye gusa.”

336. What is model monitoring, and why is it critical after deployment?

- **English:**
Model monitoring involves continuously tracking the performance, accuracy, and behavior of a deployed model in production to detect issues such as data drift, anomalies, or performance degradation.
- **Kinyarwanda:**
Model monitoring ni ugukurikirana uko model ikora mu gihe nyacyo (production) mu buryo bw’imikorere, ukuri, n’imyitwarire, kugira ngo hamenyekane ibibazo nka data drift, anomalies, cyangwa kugabanuka k’umusaruro.

337. How does A/B testing contribute to model improvement in production environments?

- **English:**
A/B testing allows comparing different model versions or configurations by splitting traffic between them, helping determine which model performs better in real-world conditions.
- **Kinyarwanda:**
A/B testing ituma habaho kugereranya versions zitandukanye za model cyangwa configurations, aho traffic igabanywa, bigafasha kubona model ikora neza mu buzima busanzwe.

338. What strategies can be implemented to detect and manage model drift over time?

- **English:**
Strategies include **continuous monitoring of performance metrics, scheduled retraining with new data, setting up alerts for significant changes, and using statistical tests to detect drift.**
- **Kinyarwanda:**
Ushobora gukoresha gukurikirana metrics buri gihe, guteganya retraining hakoreshejwe amakuru mashya, gushyiraho alert igihe habaye impinduka zikomeye, no gukoresha tests za statistique mu gutahura drift.

339. Discuss the key security challenges in deploying machine learning models and methods to mitigate them.

- **English:**
Key challenges include **unauthorized access, data breaches, adversarial attacks, and injection attacks. Mitigation methods include encryption, secure authentication (e.g., OAuth), regular security audits, and robust API security measures.**
- **Kinyarwanda:**
Ibibazo by'umutekano birimo kwinjirirwa mu buryo butemewe, kwibwa amakuru, adversarial attacks, n'ibitero bya injection. Uburyo bwo kubikumira ni ugukoresha encryption, secure authentication (nko OAuth), audits za security buri gihe, ndetse no gushyiraho uburyo bukomeye bwo kurinda API.

340. Explain the difference between batch inference and real-time inference, and provide use-case scenarios for each.

- **English:**
Batch inference processes large volumes of data at scheduled intervals and is suited for non-urgent tasks (e.g., daily report generation).
- **Real-time inference** processes data instantly as it arrives and is used for time-sensitive applications (e.g., fraud detection).
- **Kinyarwanda:**
Batch inference ikora ku makuru menshi mu gihe cyateganyijwe kandi ikaba ikwiriye ku mirimo idafite igihe gikomeye (nko gukora raporo za buri muni).
- **Real-time inference** ikora amakuru ako kanya igihe agera kandi ikoreshwa mu bikorwa bifite igihe nyacyo (nko gutahura uburiganya).

Thank you very much and I wish you more success and more achievements that how we did